TIMING CHAIN REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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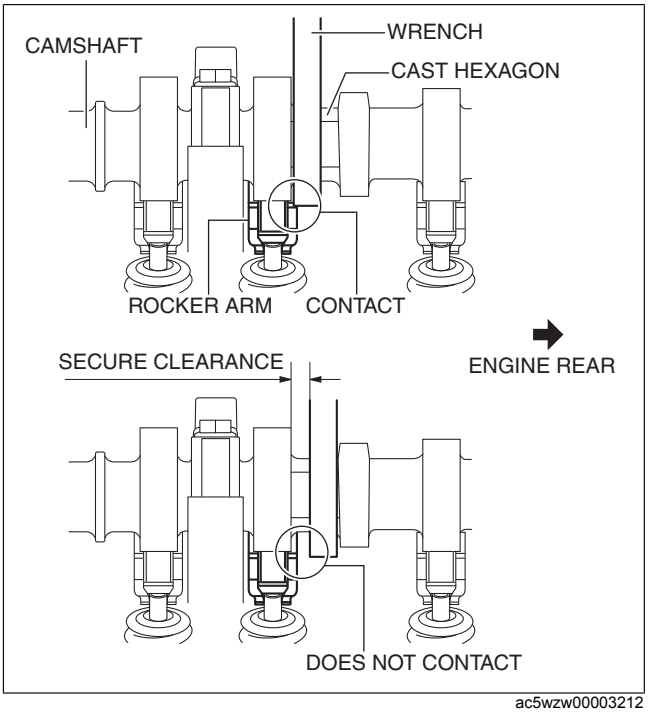
SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)

Warning

- **A hot engine can cause severe burns. Turn off the engine and wait until it is cool before servicing.**

Caution

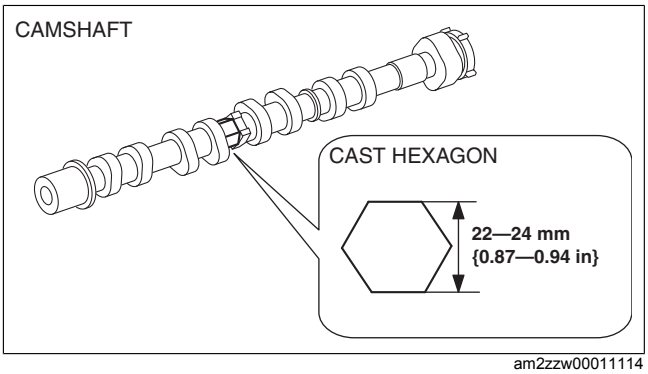
- **If the camshaft is rotated with the timing chain removed and the piston at the top dead center position, the valve may contact the piston and the engine could be damaged. When rotating the camshaft with the timing chain removed, rotate it after lowering the piston from the top dead center position.**
- **When rotating the camshaft using a wrench on the cast hexagon, the wrench may contact the rocker arm and damage the rocker arm. To prevent damage to the rocker arm when holding the camshaft on the cast hexagon, use a wrench on the rear side of the engine as shown in the figure to secure a clearance between the cam.**



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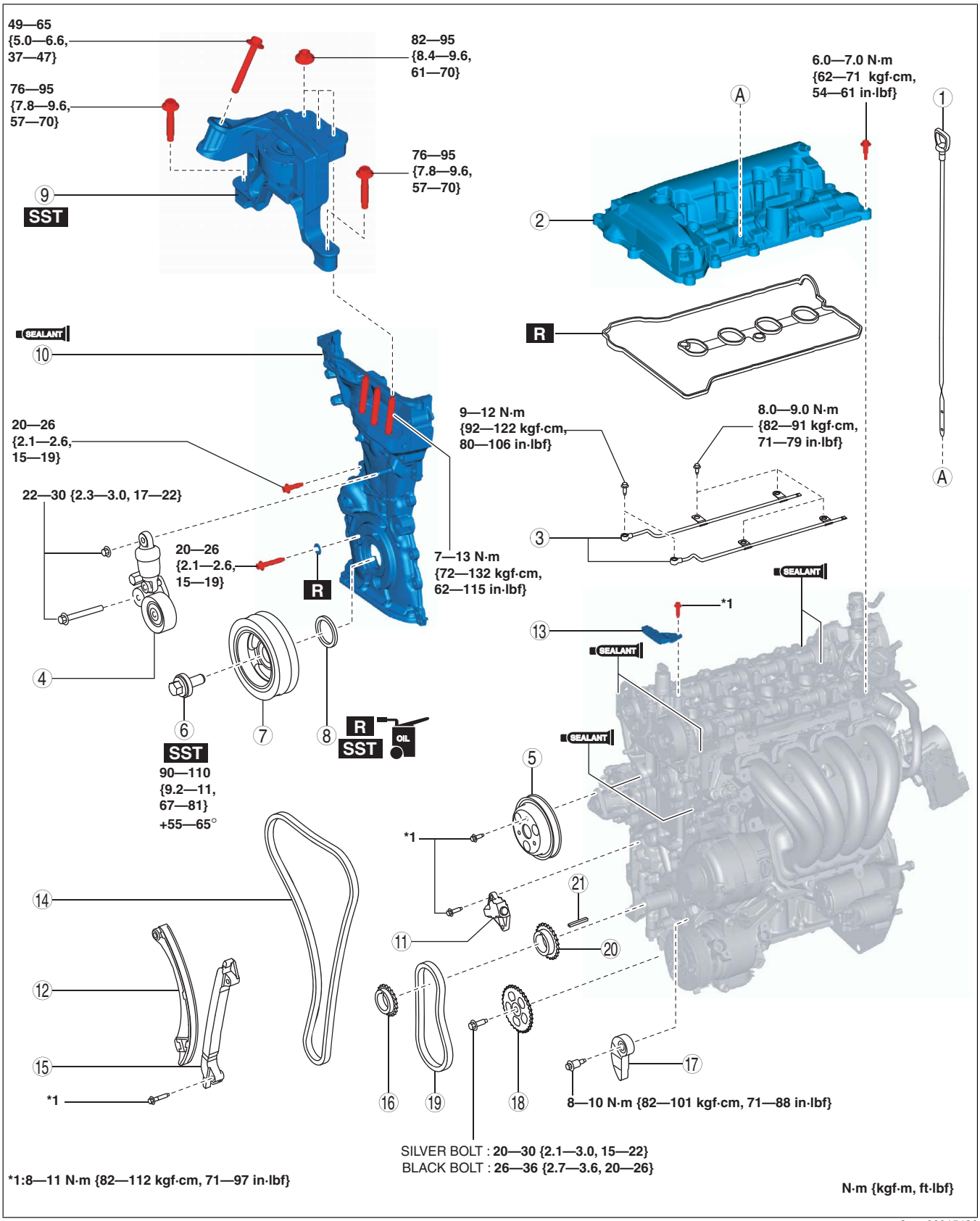
Note

- Width at the cast hexagon of the camshaft is **22—24 mm {0.87—0.94 in}**.



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1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
2. Remove the plug hole plate. (See PLUG HOLE PLATE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
3. Remove the ignition coil. (See IGNITION COIL/ION SENSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
4. Remove the front under cover No.2. (See FRONT UNDER COVER No.2 REMOVAL/INSTALLATION.)
5. Remove the splash shield (RH). (See SPLASH SHIELD REMOVAL/INSTALLATION.)
6. Remove the drive belt. (See DRIVE BELT REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
7. Drain the engine oil. (See ENGINE OIL REPLACEMENT [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
8. Remove the oil pan. (See OIL PAN REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
9. Remove in the order indicated in the table.
10. Install in the reverse order of removal.
11. Refill with the specified type and amount of the engine oil. (See ENGINE OIL REPLACEMENT [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
12. Start the engine and inspect the following:
    - Leakage of engine oil.
    - Runout and contact of pulley and belt.
    - Ignition timing, idle speed and idle mixture. (See ENGINE TUNE-UP [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)



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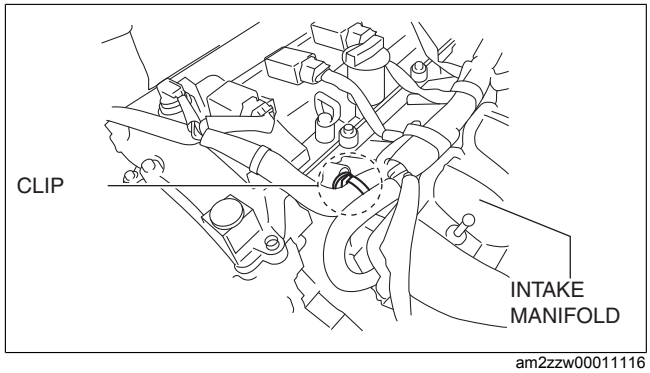
1	Dipstick
2	Cylinder head cover (See Cylinder head cover removal note.) (See Cylinder head cover installation note.)
3	Oil shower pipe (See Oil shower pipe installation note.)

4	Drive belt auto tensioner (See DRIVE BELT AUTO TENSIONER REMOVAL/ INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
5	Water pump pulley (See Water pump pulley removal note.) (See Water pump pulley installation note.)

6	Crankshaft pulley lock bolt (See Crankshaft pulley lock bolt removal note.) (See Crankshaft pulley lock bolt installation note.)
7	Crankshaft pulley
8	Front oil seal (See FRONT OIL SEAL REPLACEMENT [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
9	No.3 engine mount (See No.3 engine mount removal note.) (See No.3 engine mount installation note.)
10	Engine front cover (See Engine front cover removal note.) (See Engine front cover installation note.)
11	Chain tensioner (See Timing chain removal note.) (See Timing chain installation note.)
12	Tensioner arm (See Timing chain removal note.) (See Timing chain installation note.)
13	Chain guide (No.1) (See Timing chain removal note.) (See Timing chain installation note.)
14	Timing chain (See Timing chain removal note.) (See Timing chain installation note.)
15	Chain guide (No.2)
16	Crankshaft sprocket
17	Oil pump chain tensioner
18	Oil pump driven sprocket (See Oil pump driven sprocket removal note.) (See Oil pump driven sprocket installation note.)
19	Oil pump chain
20	Oil pump drive sprocket
21	Key

Cylinder head cover removal note

1. Disconnect the OCV connector.
2. Remove the wiring harness clips secured to the cylinder head cover.
3. Remove the wiring harness clip secured to the intake manifold as shown in the figure, and set the wiring harness aside.
4. Disconnect the ventilation hose.
5. Disconnect the brake vacuum hose from the intake manifold and set it aside.
6. Remove the cylinder head cover.



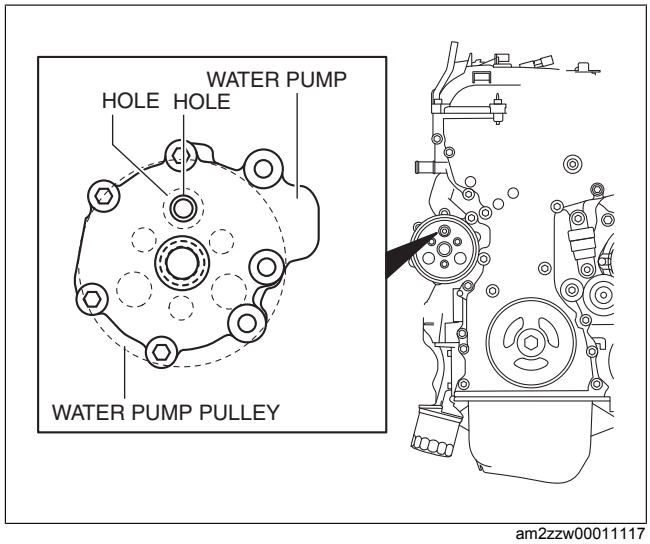
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Water pump pulley removal note

Caution

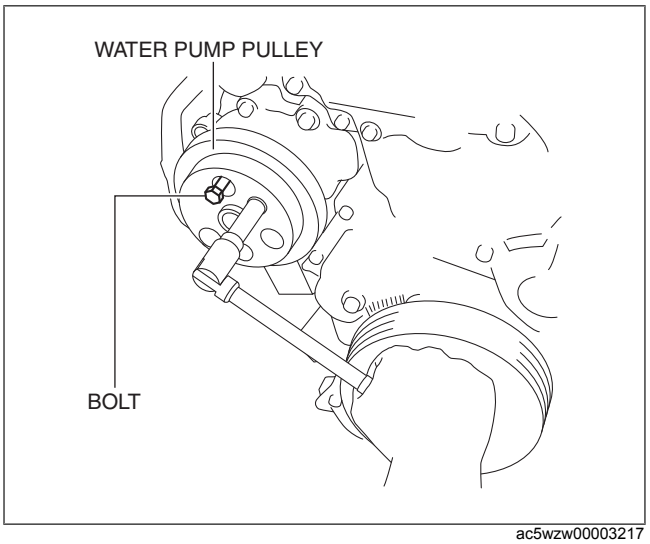
- **Be careful not to damage the belt groove and surface of the water pump pulley when using tools, otherwise it will cause wear, breakage, abnormal noise of the drive belt (stretch belt), damage to the pulley, and rust.**

1. Align the water pump pulley hole with the water pump hole as shown in the figure.



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2. Insert an appropriate bolt (length **approx. 70 mm {2.8 in}**) into the water pump hole as shown in the figure, and lock the water pump pulley against rotation.
3. Remove the water pump pulley.
4. Remove the bolt used for locking the water pump pulley against rotation.

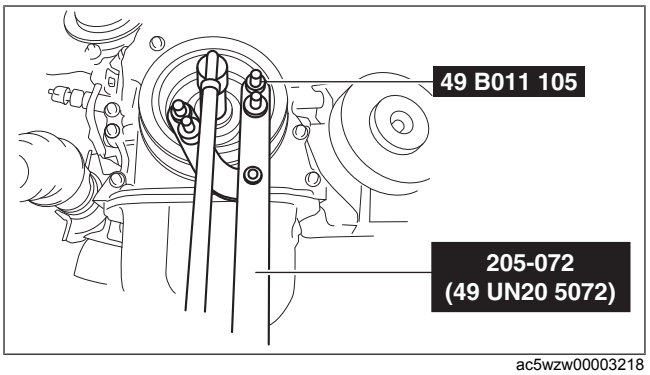


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Crankshaft pulley lock bolt removal note

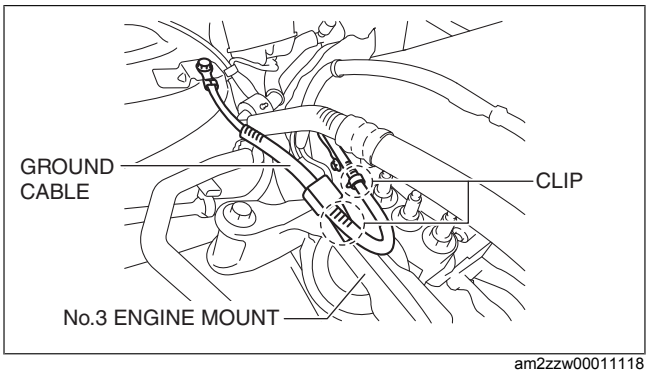
Caution
- **To prevent damage to the crankshaft pulley, protect the crankshaft pulley with a clean cloth so that the bar part (205-072) of the SST does not contact the crankshaft pulley.**
- **To prevent damage to the engine front cover, be careful not to allow the tab part (49 B011 105) of the SST to get caught between the crankshaft pulley and the engine front cover.**

1. Install the **SST** to the crankshaft pulley, and lock the crankshaft against rotation.
2. Remove the crankshaft pulley lock bolt.



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No.3 engine mount removal note
1. Remove the clips shown in the figure and set the ground cable aside.

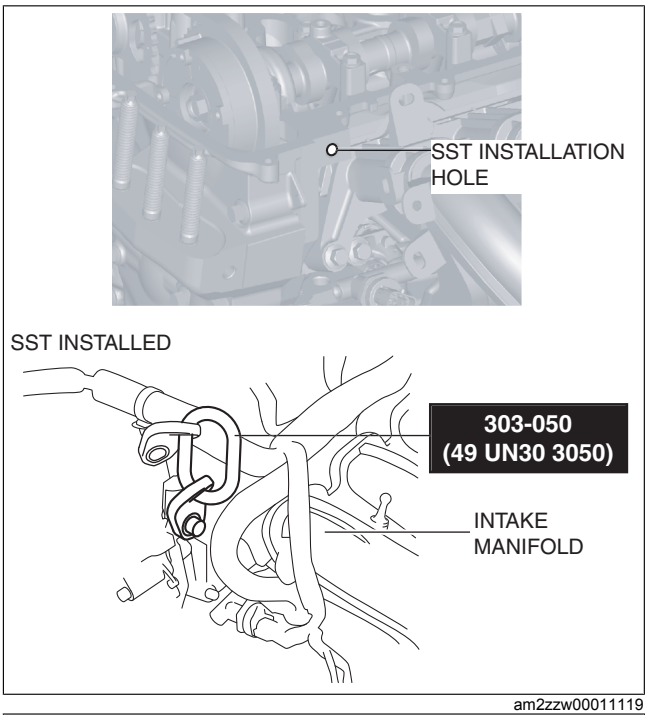


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2. Install the **SST** and the following bolt to the position shown in the figure.

   Bolt:
   part number 99794 1025 or an M10 × 1.25, length 25 mm {0.98 in}
   Tightening torque
   38—51 N·m {3.9—5.2 kgf·m, 29—37 ft·lbf}

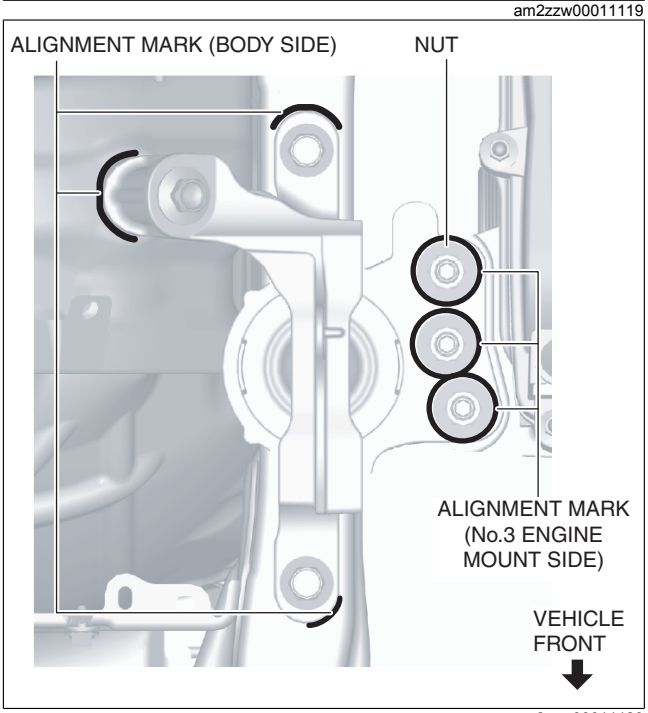
   Caution
   - **Slots have been adopted for the No.3 engine mount installation holes. If the No.3 engine mount is deviated from the original position when installing the No.3 engine mount, engine noise or vibration could increase. Before removing the No.3 engine mount, place alignment marks on the No.3 engine mount and body so that they can be assembled to the same positions as before removal.**



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3. Place alignment marks on the locations shown in the figure so that they can be assembled to the same positions as before removal.

   Note
   - Paint so that the No.3 engine mount is framed on the body side and the outline of the nut is framed on the No.3 engine mount side.



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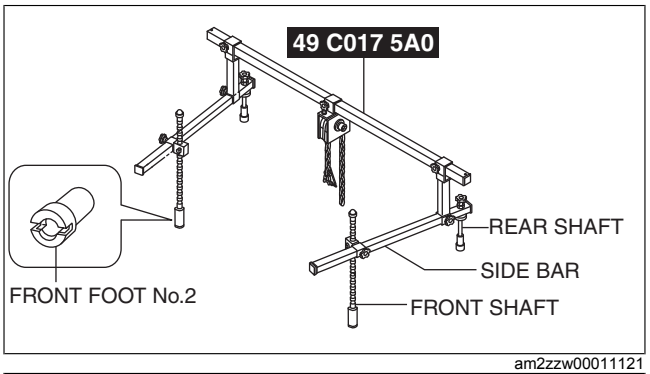
4. Install the **SST** using the following procedures.

   Caution
   - **Refer to the SST instruction manual for the basic handing procedure.**

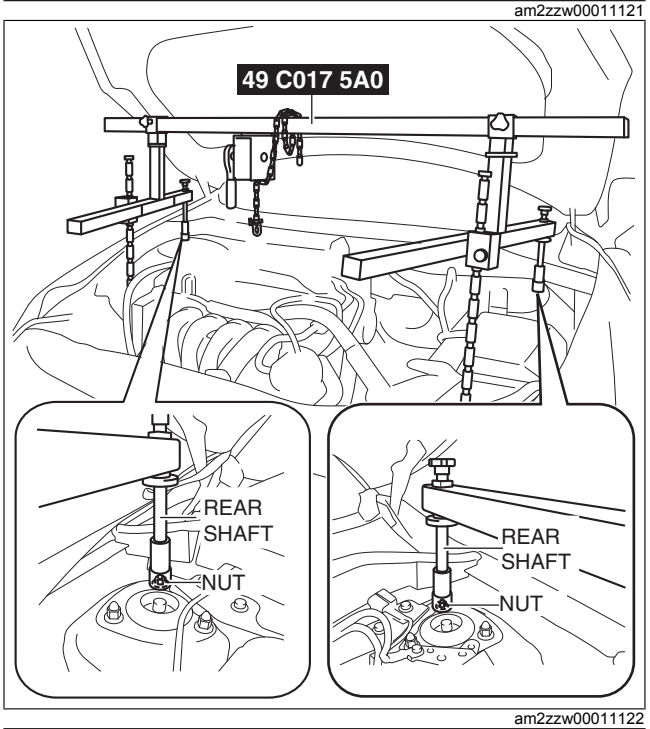
Note

- Install front foot No. 2 to the left and right front shafts of the **SST**.

(1) Set the rear shafts of the **SST (49 C017 5A0)** on the left/right front shock absorber nuts (top) as shown in the figure.

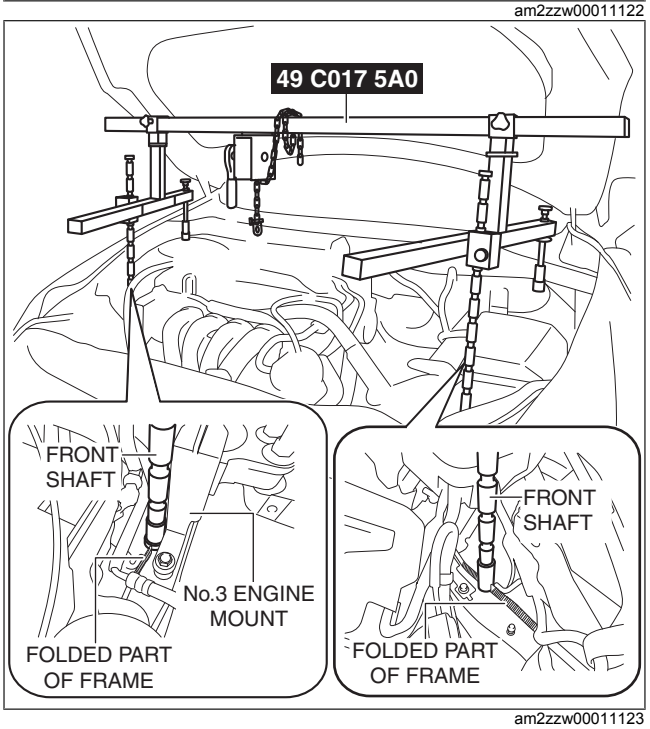


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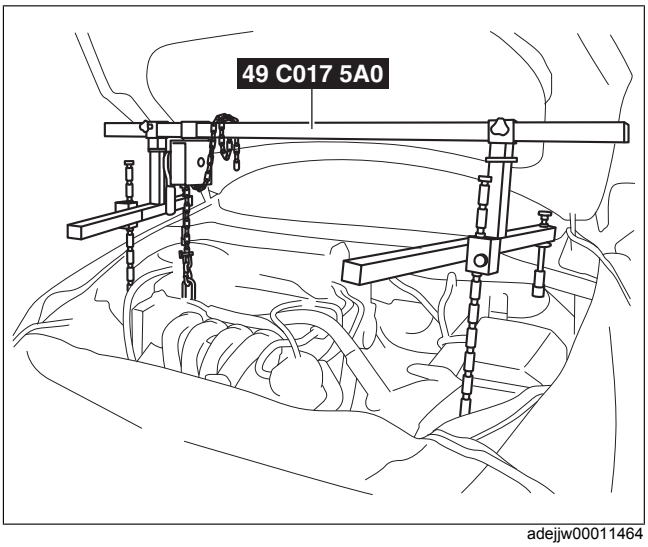
(2) Set the front shafts of the **SST (49 C017 5A0)** on the bent up part of the frame as shown in the figure.



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(3) Adjust the height of the left and right side bars of the **SST** so that they are leveled, then tighten each part.
(4) Apply tension to the chain to hold the engine.

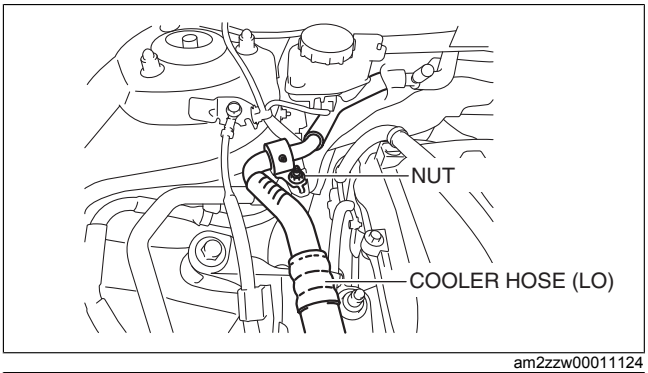
5. Remove the No.3 engine mount.



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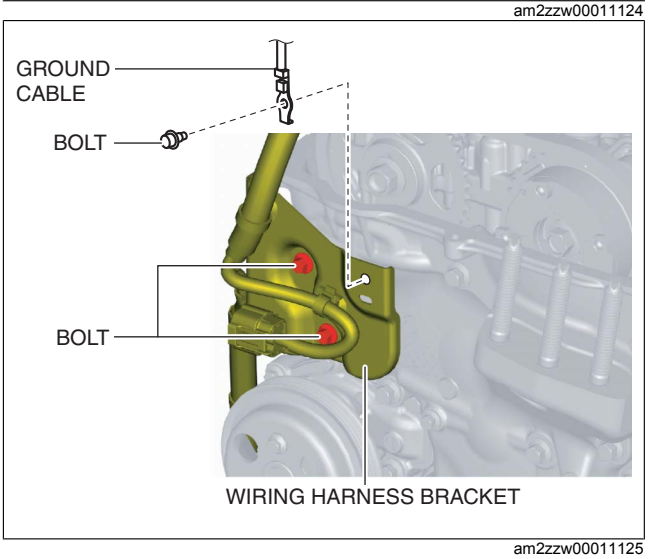
Engine front cover removal note

1. Remove the nut shown in the figure and set the cooler hose (LO) aside.



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2. Set the ground cable and wiring harness bracket shown in the figure aside.

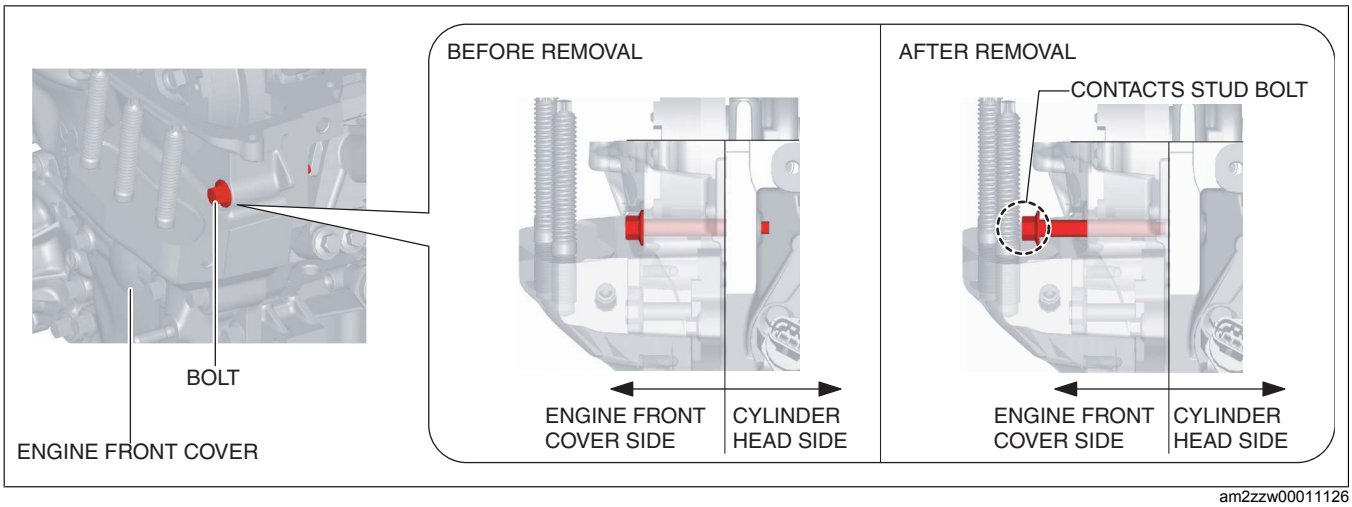


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3. Remove the engine front cover installation bolts.

Note

- Because the bolt shown in the figure contacts the stud bolt when the bolt is loosened, the bolt cannot be completely removed. However, the bolt comes out of the cylinder head so that the engine front cover can be removed.

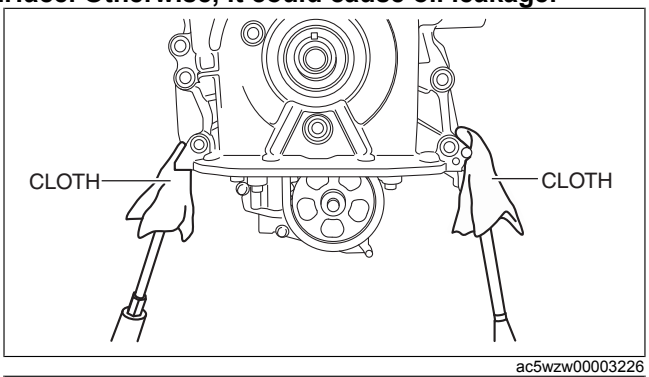


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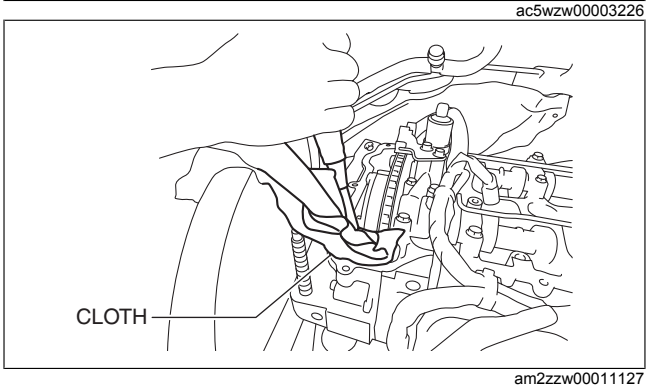
4. Using a screwdriver wrapped in a cloth, peel the sealant away a little at a time, and remove the engine front cover.

Caution

- **Do not apply excessive force to the screwdriver. Otherwise, the engine front cover could be damaged.**
- **Be careful not to scratch or damage the seal surface. Otherwise, it could cause oil leakage.**



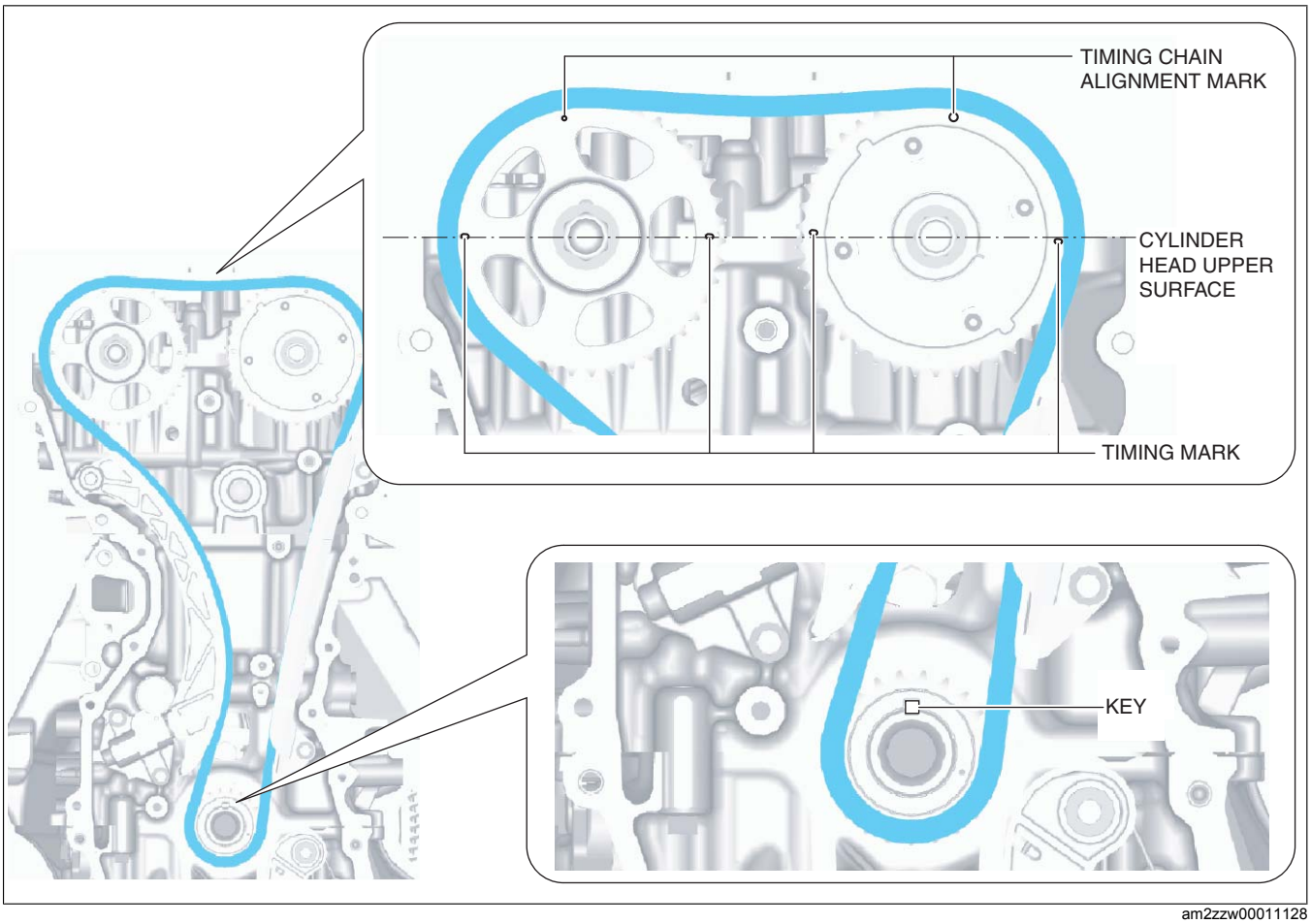
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Timing chain removal note

1. Rotate the crankshaft clockwise to align the timing marks and the key position as shown in the figure, and set cylinder No.1 at top dead center (TDC).

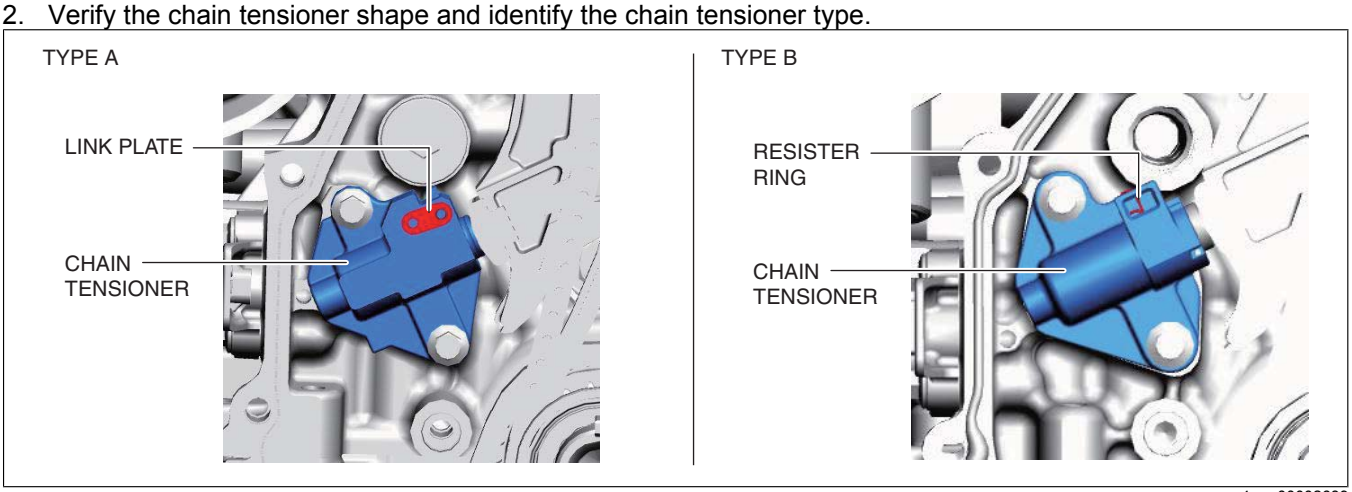


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Note

- The timing mark is not completely parallel with the upper surface of the cylinder head.

2. Verify the chain tensioner shape and identify the chain tensioner type.

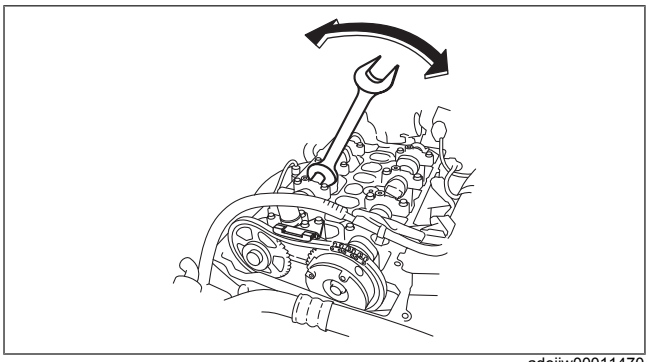


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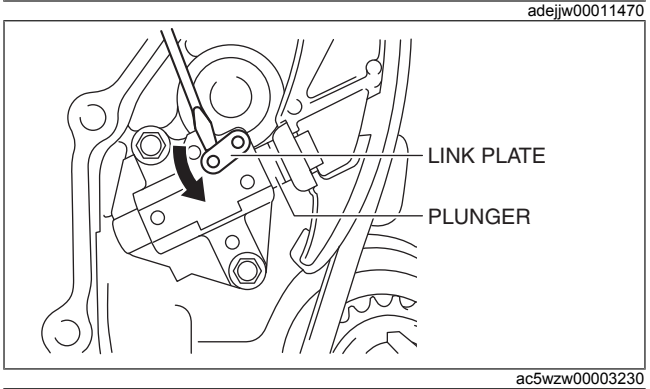
Note

- Verifying the chain tensioner type is necessary because the following procedures may differ according to the type of tensioner.

3. While moving the exhaust camshaft back and forth in the direction of the arrow using a wrench on the cast hexagon, press down the link plate of the timing chain tensioner using a precision screwdriver and release the plunger lock. (Chain tensioner (type A))

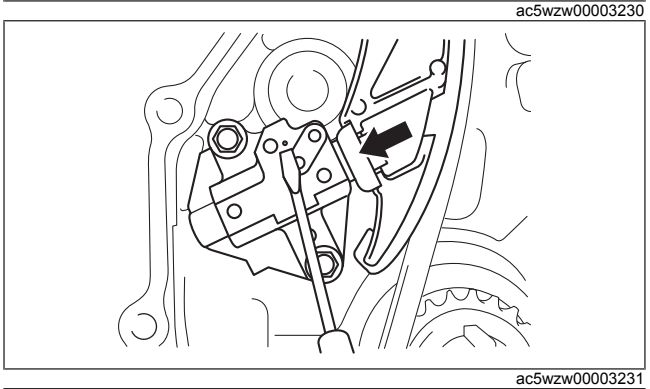

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Note
- When moving the exhaust camshaft back and forth, the timing chain pushes the plunger in the chain tensioner making it easier to operate the link plate.

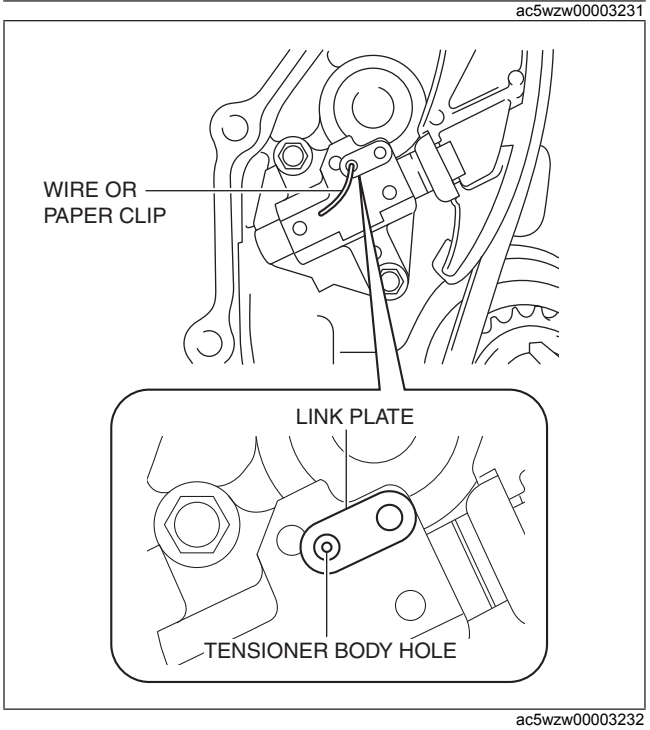


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4. Push back the plunger slowly in the direction shown in the figure with the link plate still pushed down. (Chain tensioner (type A))
5. Remove the screwdriver from the link plate with the plunger still pushed down. (Chain tensioner (type A))
6. Release the force slightly from the plunger, and move it back and forth **2—3 mm {0.08—0.11 in}**. (Chain tensioner (type A))


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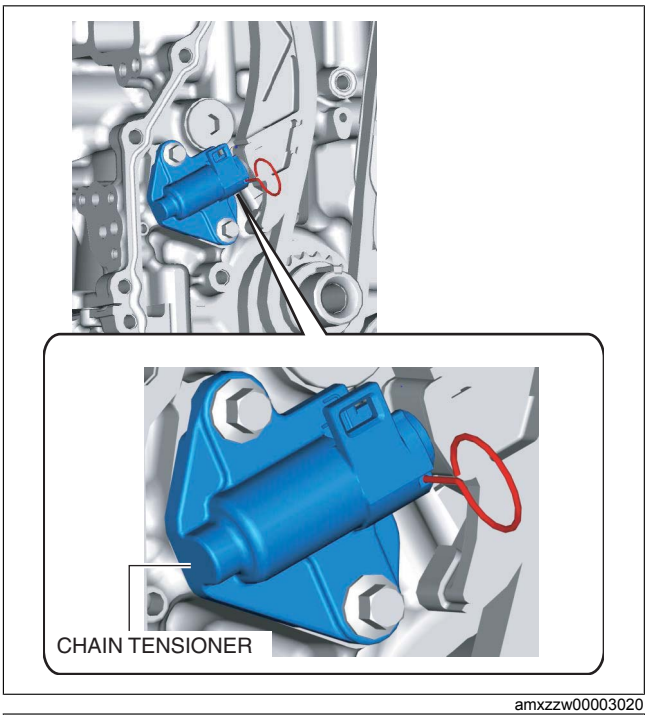
7. Insert a wire with an **approx. diameter of 1.5 mm {0.059 in}** or a paper clip where the link plate hole and the tensioner body hole overlap to secure the link plate and lock the plunger. (Chain tensioner (type A))



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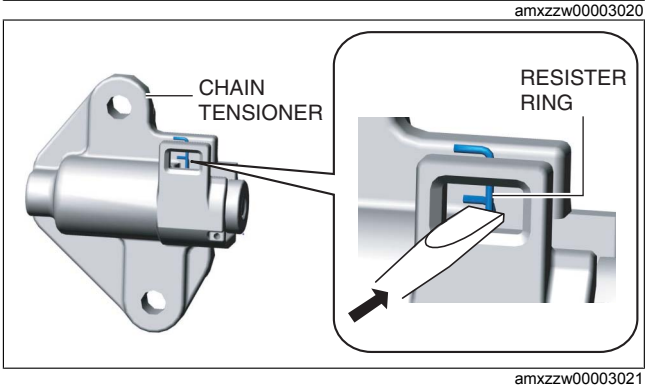
8. Loosen the chain tensioner using the following procedure: (Chain tensioner (type B))

(1) Insert a metal rod (diameter **approx. 1.4 mm {0.055 in}**, length **approx. 55 mm {2.2 in}**) into the hole in the body of the chain tensioner.



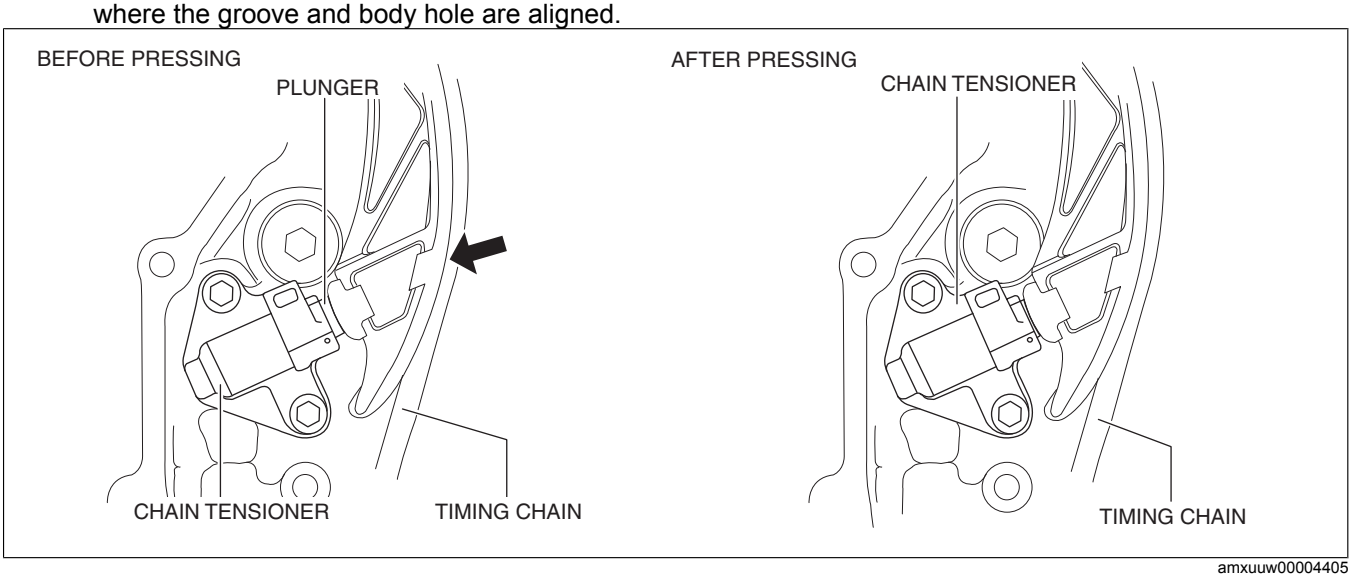
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(2) Press the resister ring and unlock the plunger.



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(3) Press the timing chain in the direction of the arrow and press in the chain tensioner plunger to the position where the groove and body hole are aligned.



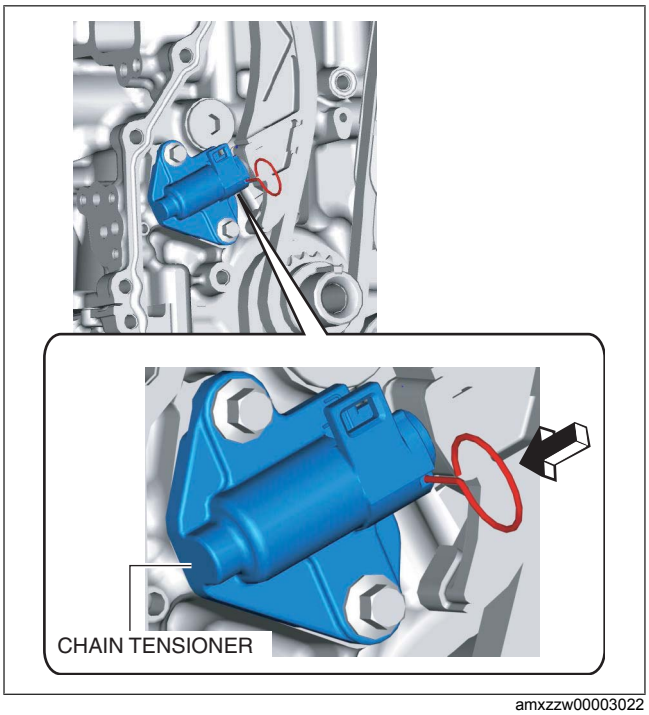
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(4) With the plunger pressed in, further insert the metal rod set in (1) above.

Note
- The rod secures the plunger and releases the tension.

(5) Loosen the power of the hand pressing the plunger and verify that the pressed-in rod does not move.

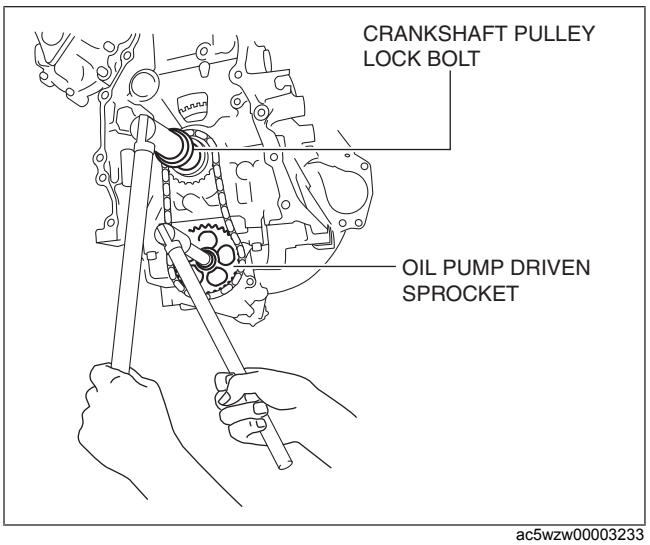
9. Remove the chain tensioner, tensioner arm, and the chain guide (No.1).
10. Remove the timing chain.



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Oil pump driven sprocket removal note

1. Temporarily assemble the crankshaft pulley lock bolt, and lock the oil pump against rotation as shown in the figure.
2. Remove the oil pump driven sprocket.
3. Remove the temporarily assembled crankshaft pulley lock bolt.



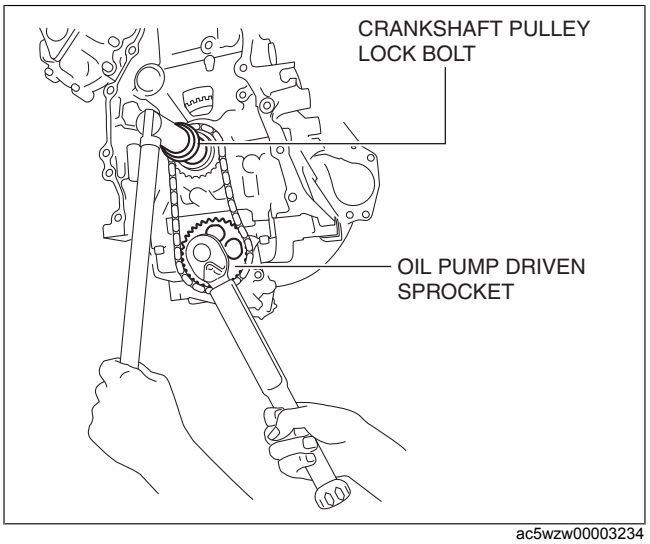
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Oil pump driven sprocket installation note

1. Temporarily assemble the crankshaft pulley lock bolt, and lock the oil pump against rotation as shown in the figure.
2. Install the oil pump driven sprocket.

Tightening torque
Silver bolt: 20—30 N·m {2.1—3.0 kgf·m, 15—22 ft·lbf}
Black bolt: 26—36 N·m {2.7—3.6 kgf·m, 20—26 ft·lbf}

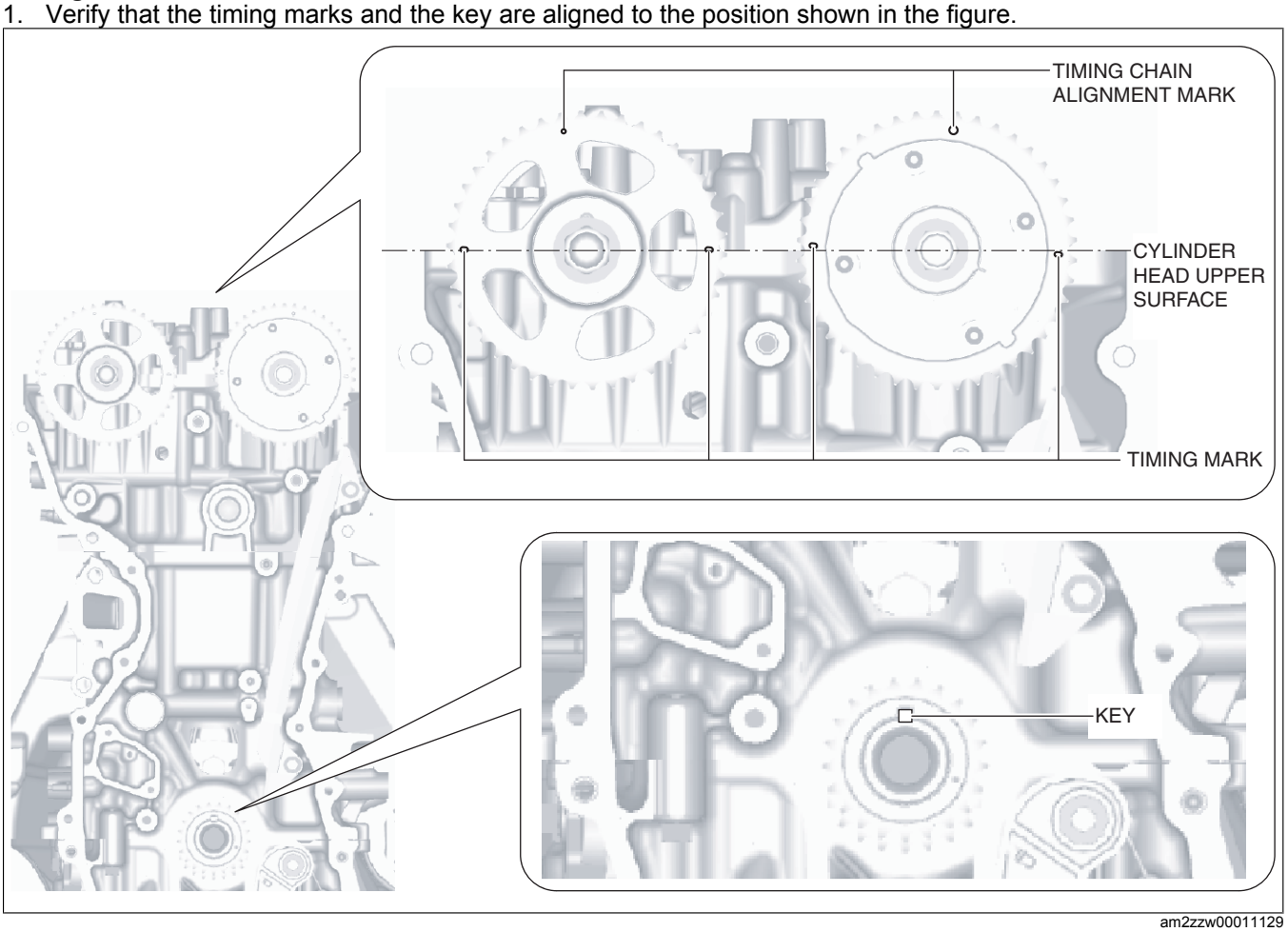
3. Remove the temporarily assembled crankshaft pulley lock bolt.



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Timing chain installation note

1. Verify that the timing marks and the key are aligned to the position shown in the figure.



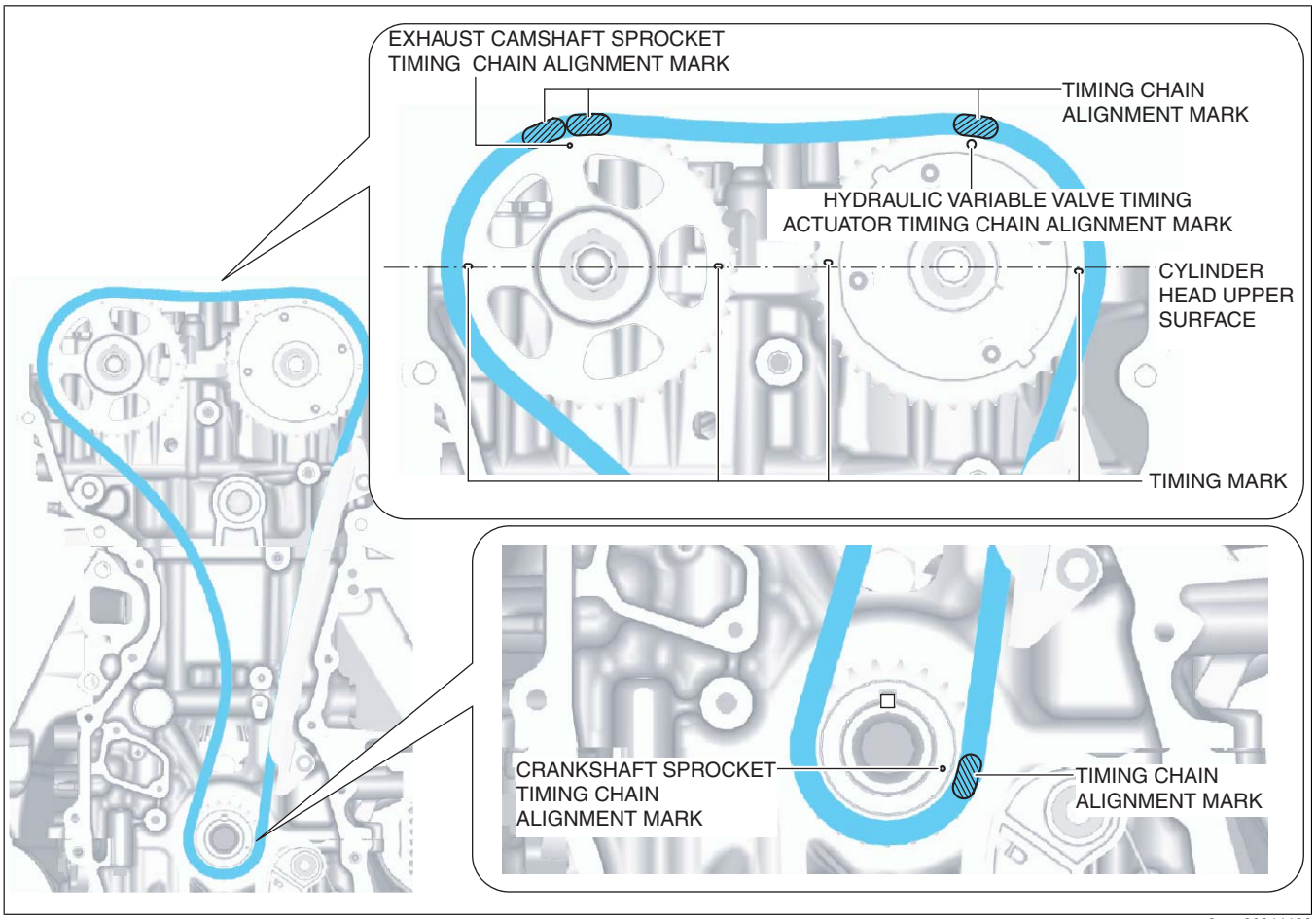
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Note

- The timing mark is not completely parallel with the upper surface of the cylinder head.

- If they are not in the position shown in the figure, rotate the camshaft and crankshaft to set the cylinder No. 1 top dead center (TDC).

2. Install the timing chain while aligning the marks on the timing chain as shown in the figure.



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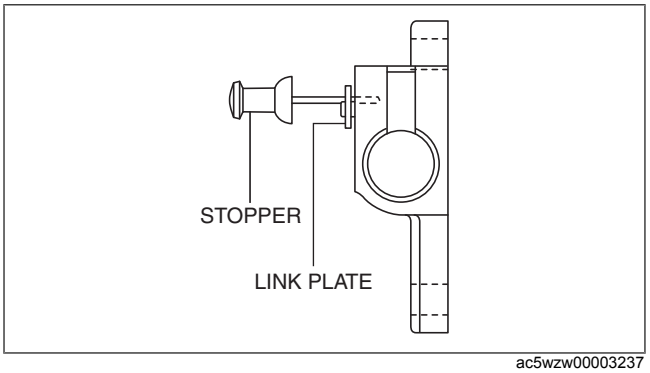
3. Install the chain guide (No.1).

   Tightening torque
   8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}

4. Install the tensioner arm.
5. Install the chain tensioner.

   Tightening torque
   8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}

6. After installing the chain tensioner, remove the installed wire or paper clip, and then apply tension to the timing chain. (Chain tensioner (type A))
   - If a new chain tensioner is used, remove the installed stopper.

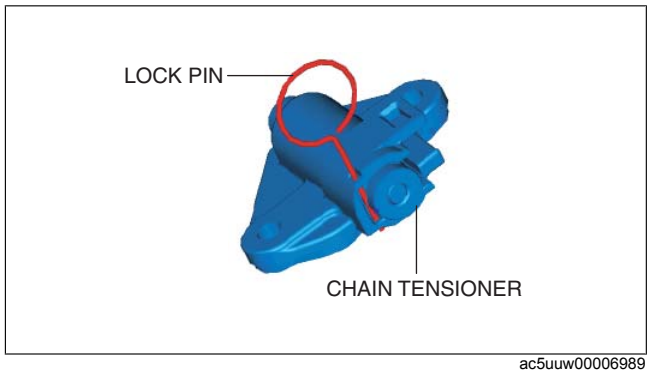


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7. After installing the timing chain tensioner, remove the installed rod, and then apply tension to the timing chain. (Chain tensioner (type B))

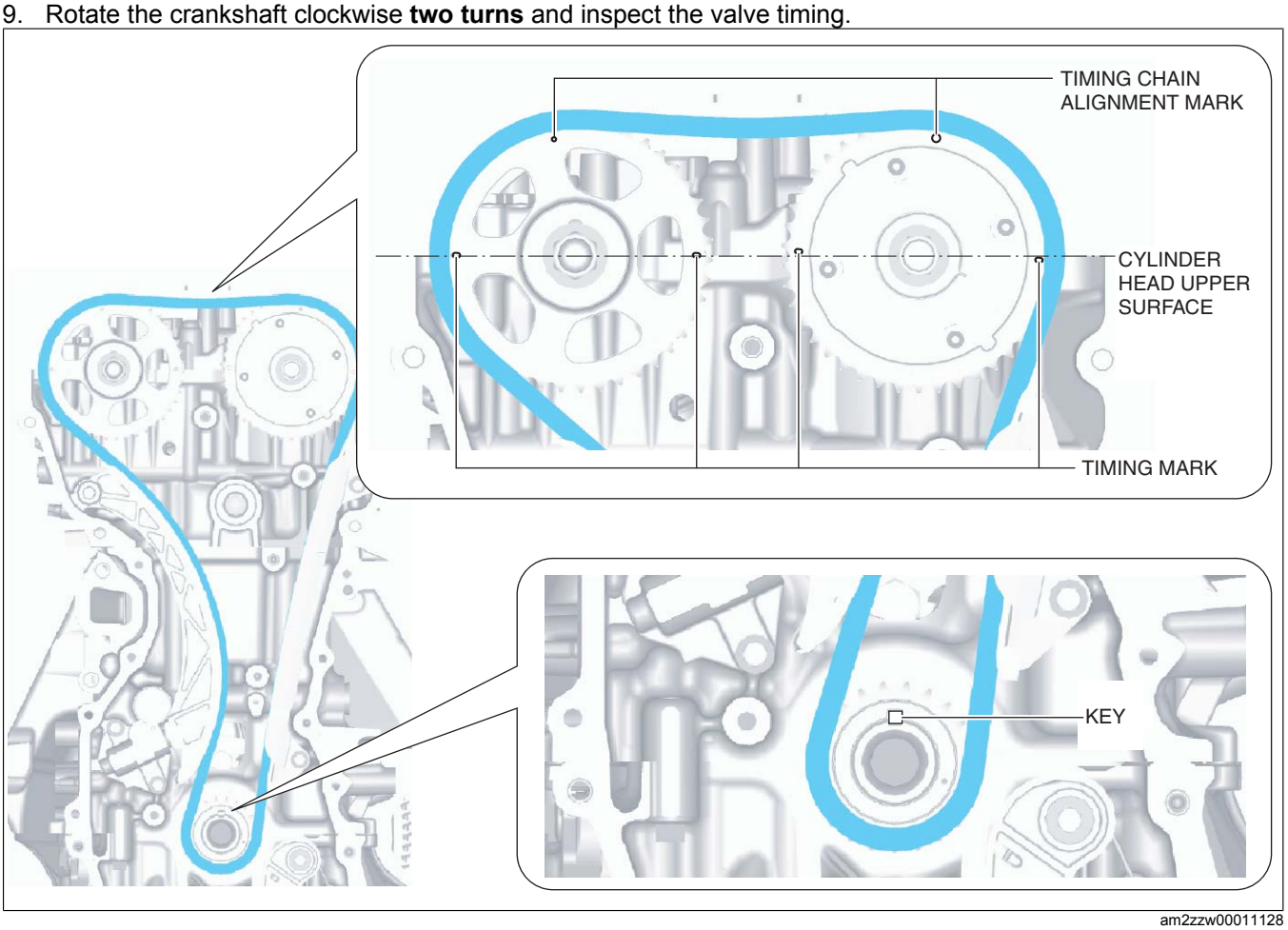
- If a new chain tensioner is used, remove the installed lock pin.

8. Verify that there is no looseness in the timing chain, and re-verify that each sprocket is in the specified location.



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9. Rotate the crankshaft clockwise **two turns** and inspect the valve timing.



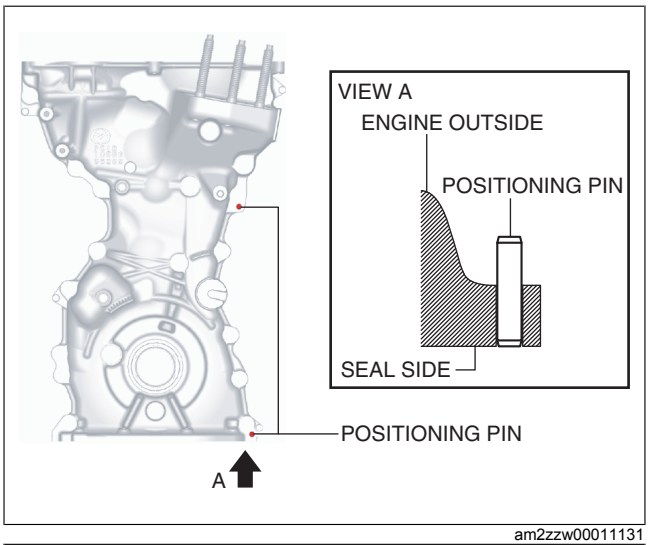
am2zzw00011128

Note

- The timing mark is not completely parallel with the upper surface of the cylinder head.

Engine front cover installation note

Note
- For a new engine front cover, the positioning pins in the **two locations** shown in the figure project to the outside of the engine.



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1. If the engine front cover is newly replaced, tap the positioning pins in the **two locations** to the seal surface side.

   Caution
   - **If a bolt with sealant adhering to it is used, it could result in cracks in the cylinder head and cylinder block.**

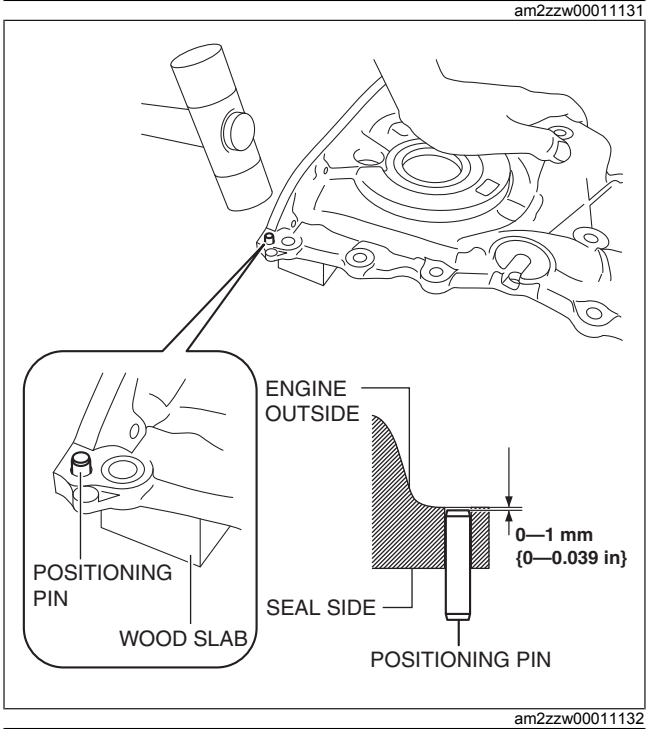
2. When reusing an engine front cover installation bolts, remove sealant adhering to the bolts.

   Caution
   - **If oil, dirt and sealant remains on the sealant application area, the silicone sealant will not seal which will cause oil leakage.**

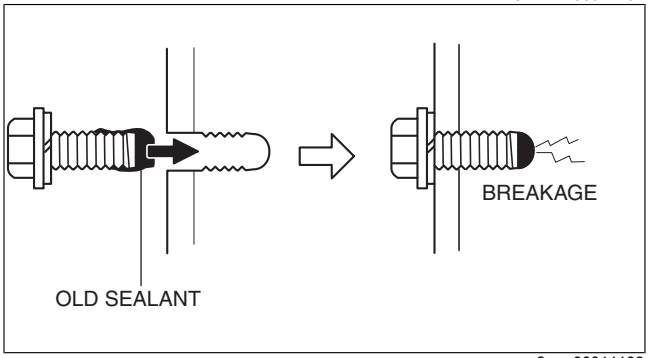
3. Completely clean and remove oil, dirt, sealant or other foreign matter that may be adhering to the engine front cover, cylinder head, and cylinder block.

   Caution
   - **Apply the silicon sealant in a single, unbroken line.**
   - **To prevent silicone sealant from hardening, adhere the engine front cover and the cylinder block firmly within 10 min. after applying silicone sealant. After adhering them, tighten the installation bolts immediately.**

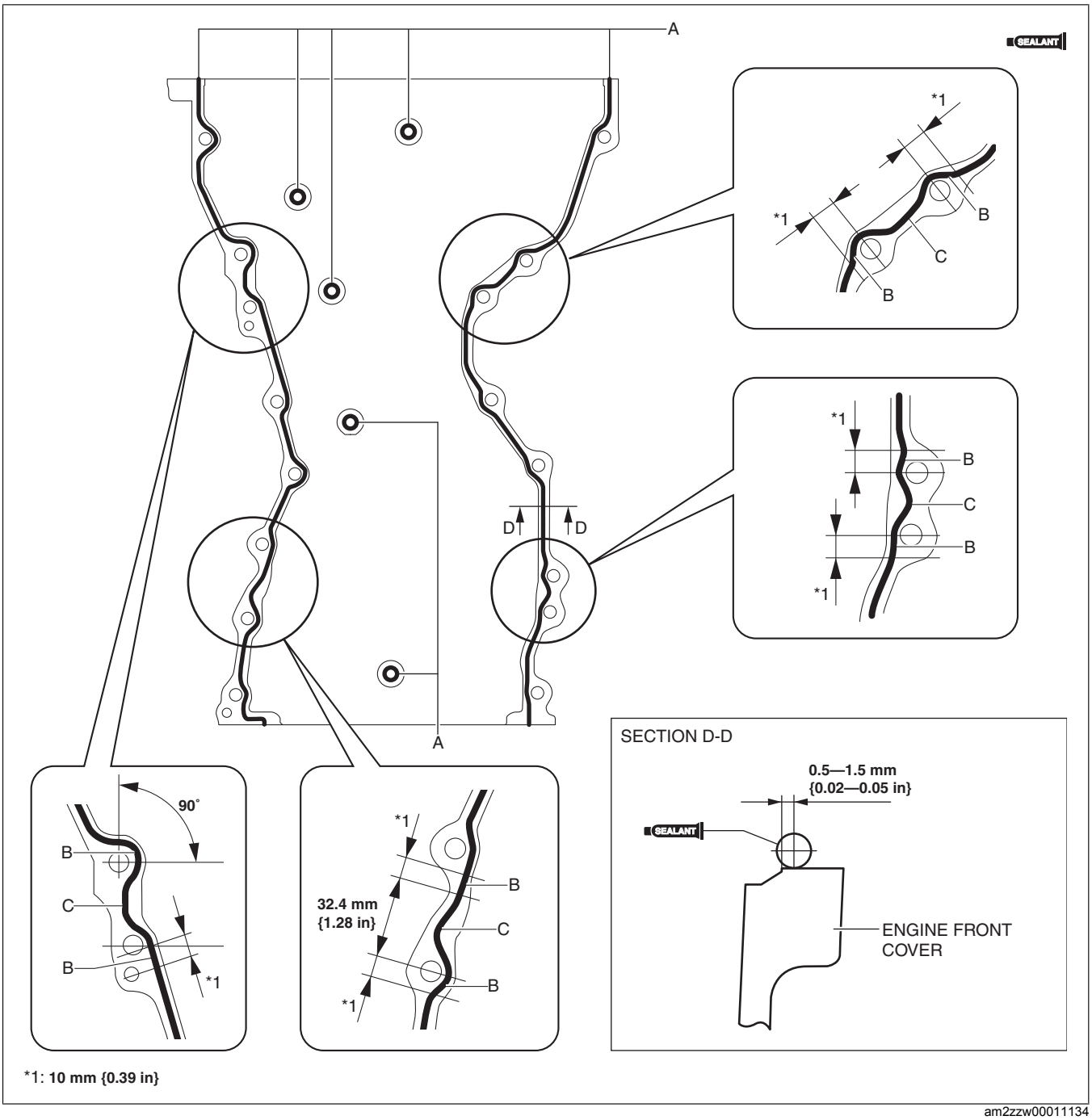
4. Apply silicone sealant (TB1217D or equivalent) to the engine front cover as shown in the figure.



am2zzw00011132



am2zzw00011133



am2zzw00011134

Bead thickness

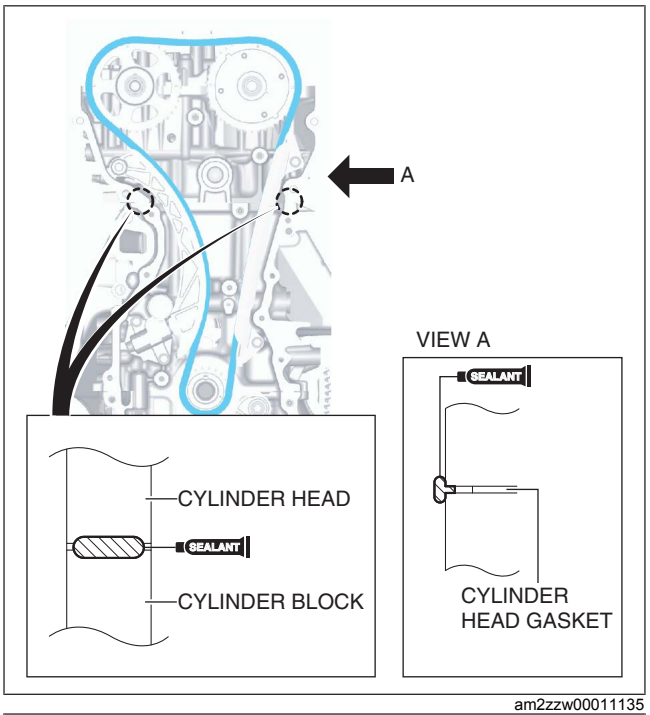
A: 2—6 mm {0.1—0.2 in}
B: 4—6 mm {0.16—0.23 in}
C: 4—8 mm {0.2—0.3 in}

5. Apply silicone sealant (TB1217D or equivalent) to the areas shown in the figure.

   Caution
   - **Apply the silicone sealant so that it goes into the cylinder head gasket.**

6. Install the engine front cover to the engine.

   Note
   - Temporarily install an appropriate bolt to the drive belt auto tensioner installation bolt hole to prevent:
     — A silicone sealant adhesion malfunction in the drive belt auto tensioner installation bolt hole.
     — A bolt mis-installation due to silicone sealant hardening.

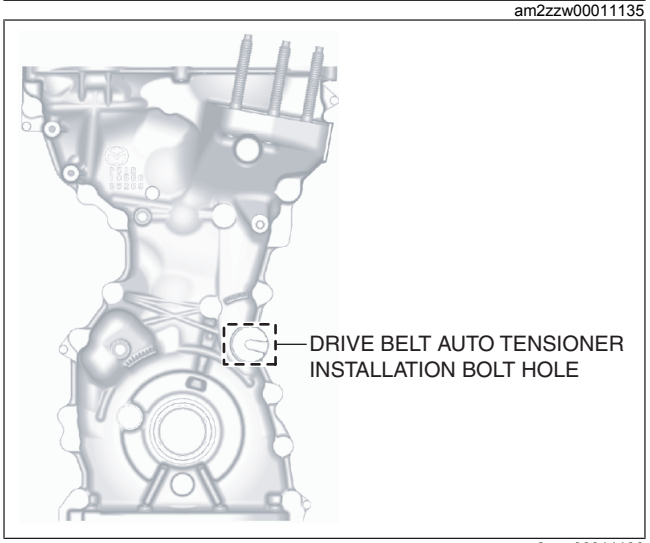


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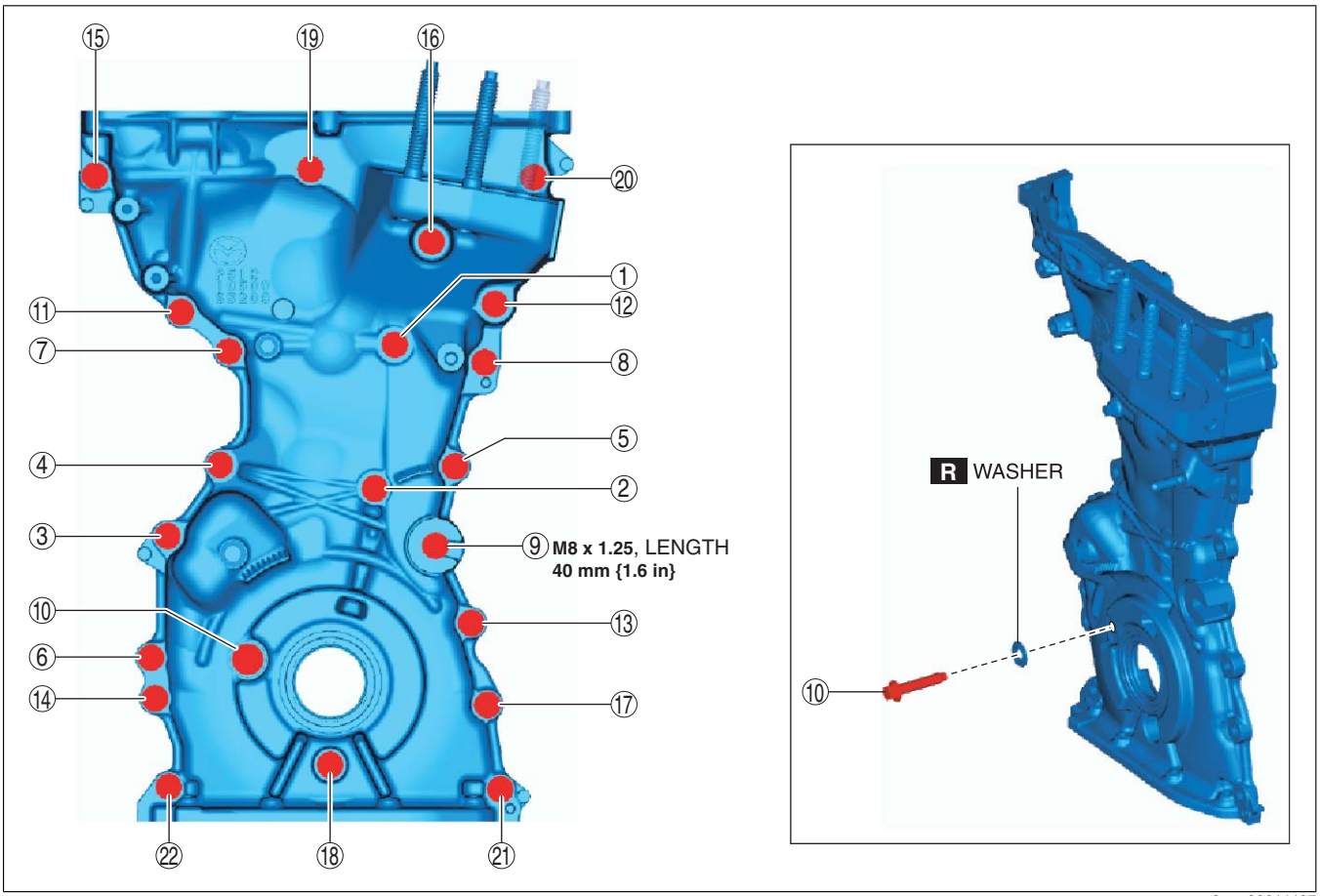
7. Prepare an appropriate **M8 × 1.25** bolt (length **40 mm {1.6 in}**).

   Caution
   - **For the number 10 bolt of the tightening order, install the bolt with new washer.**

8. Tighten the engine front cover installation bolts in the order shown in the figure.



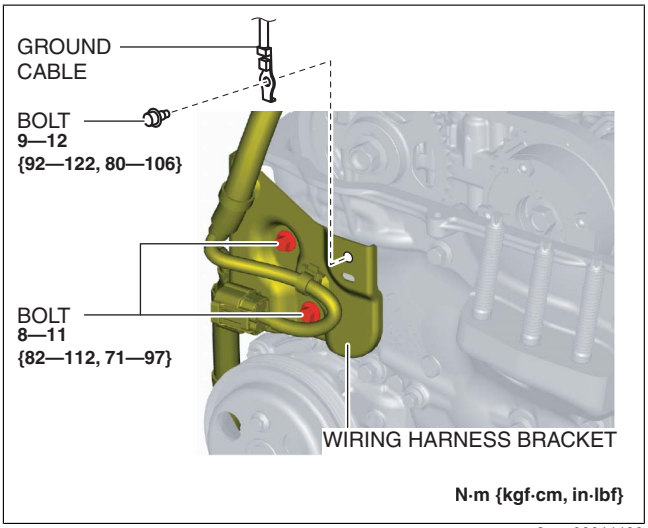
am2zzw00011136



am2zzw00011137

Tightening torque
20—26 N·m {2.1—2.6 kgf·m, 15—19 ft·lbf}

9. Remove the bolt installed to the drive belt auto tensioner installation bolt hole when installing the drive belt auto tensioner.
10. Install the ground cable and wiring harness bracket shown in the figure.



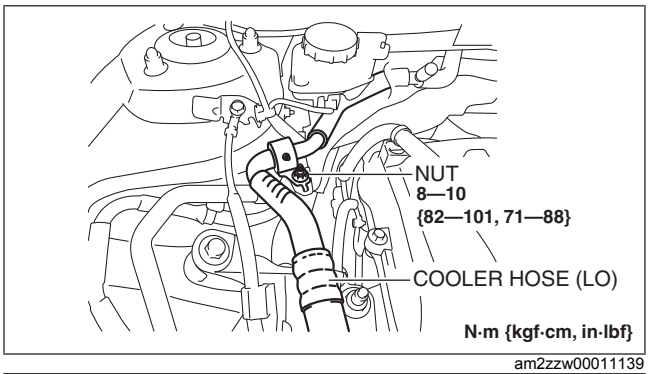
am2zzw00011138

11. Install the cooler hose (LO).

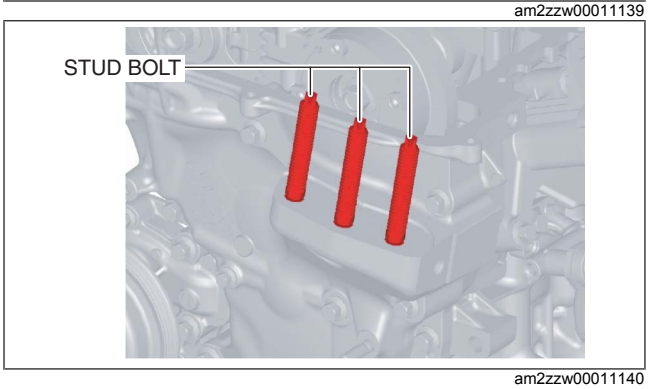
Note
- Install stud bolts because there are no stud bolts installed on a new engine front cover.
- If the engine front cover is reused, tighten the stud bolts because they may have loosened.

12. Tighten the engine front cover stud bolts.

Tightening torque
7—13 N·m {72—132 kgf·cm, 62—115 in·lbf}



am2zzw00011139

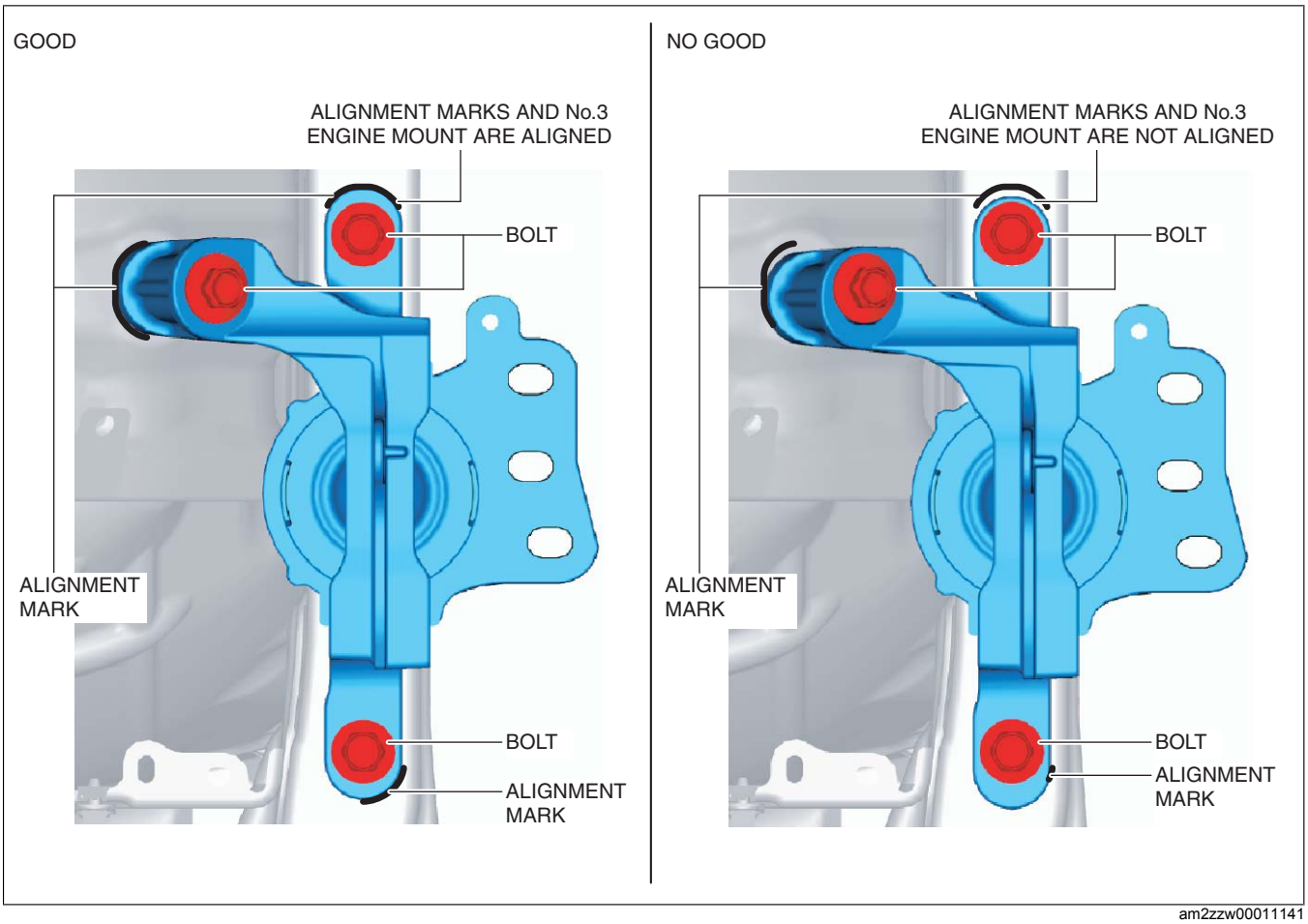


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No.3 engine mount installation note

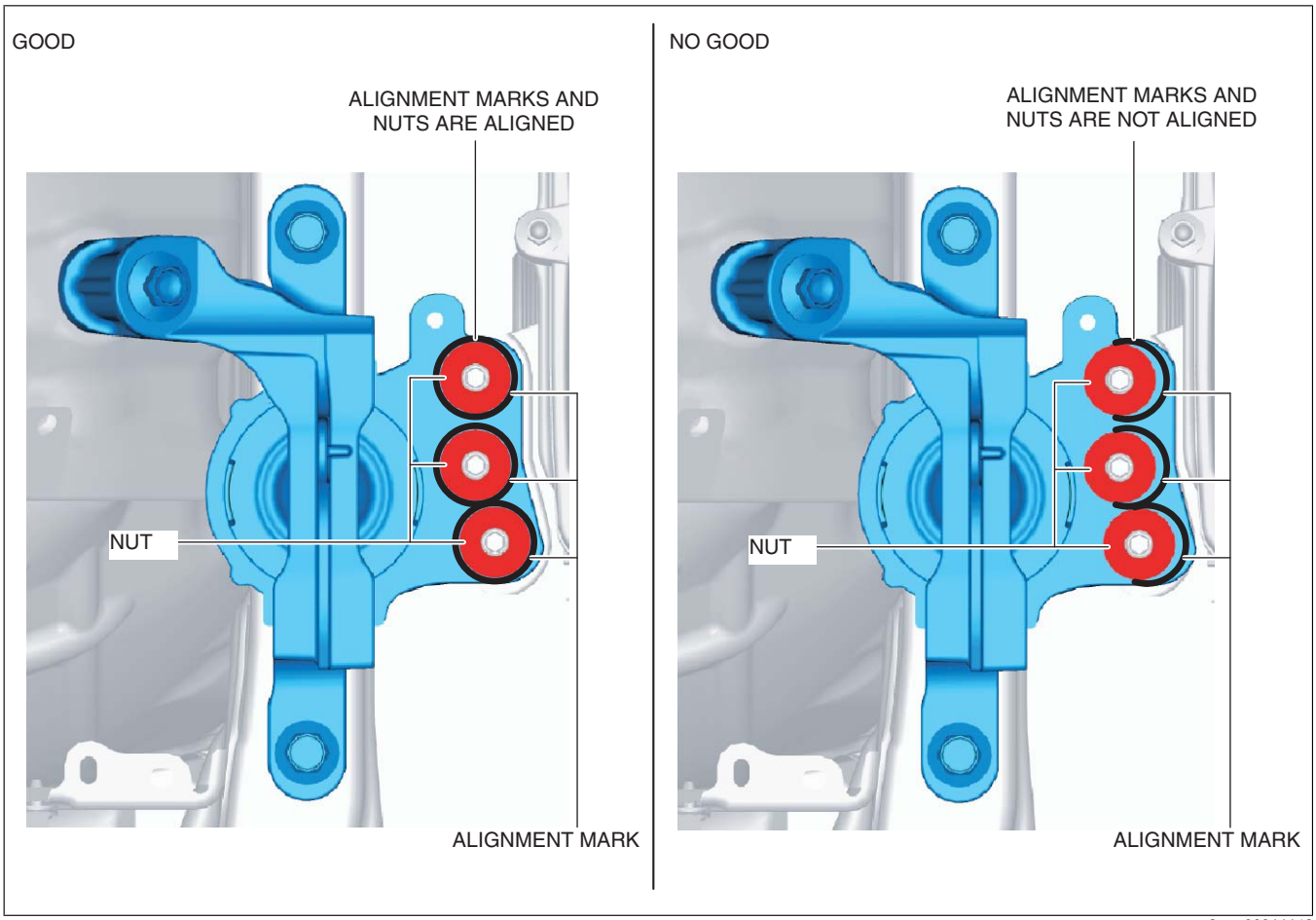
Caution
- **If the No.3 engine mount is deviated from the original position when installing the No.3 engine mount, engine noise or vibration could increase. When installing the No.3 engine mount, align the alignment mark placed during removal and install it to the original position.**

1. Temporarily tighten the No.3 engine mount installation bolts and nuts using the following procedure:
   (1) Align the alignment marks on the body side and No.3 engine mount, and temporarily tighten the bolts shown in the figure.



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(2) Temporarily tighten the nuts shown in the figure while aligning the alignment marks of the No.3 engine mount and nuts.



am2zzw00011142

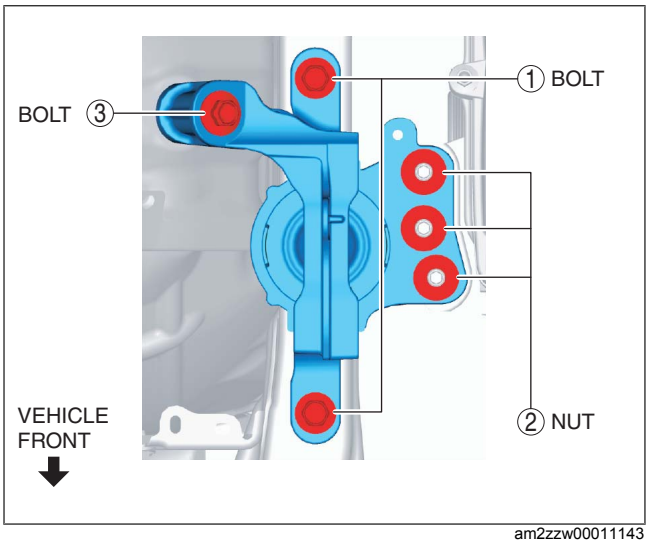
Note

- If the alignment marks are not aligned, align the alignment marks while slightly moving the engine and temporarily tighten the nuts.

2. Tighten the No.3 engine mount installation bolts and nuts in the order as shown in the figure.

Tightening torque

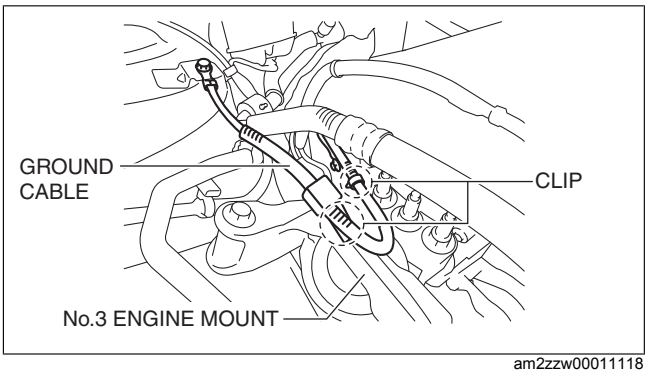
Installation position	Tightening torque
1	76—95 N·m {7.8—9.6 kgf·m, 57—70 ft·lbf}
2	82—95 N·m {8.4—9.6 kgf·m, 61—70 ft·lbf}
3	49—65 N·m {5.0—6.6 kgf·m, 37—47 ft·lbf}



am2zzw00011143

3. Remove the **SST**.

4. Install the clip of ground cable to the No.3 engine mount.



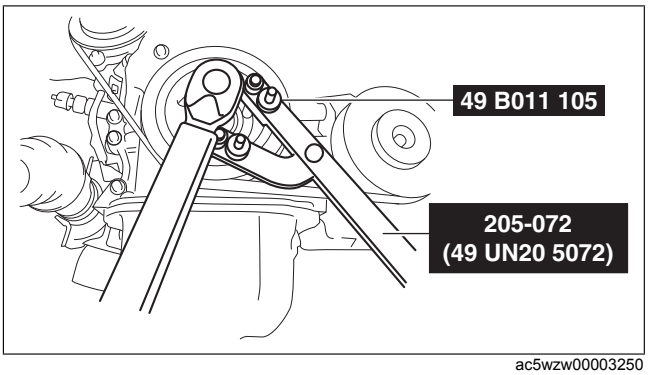
am2zzw00011118

Crankshaft pulley lock bolt installation note

Caution
- **To prevent damage to the crankshaft pulley, protect the crankshaft pulley with a clean cloth so that the bar part (205-072) of the SST does not contact the crankshaft pulley.**
- **To prevent damage to the engine front cover, be careful not to allow the tab part (49 B011 105) of the SST to get caught between the crankshaft pulley and the engine front cover.**

1. Install the **SST** on the crankshaft pulley to lock the crankshaft against rotation.
2. Tighten the crankshaft pulley lock bolt in the order shown in the following **two steps**.

Tightening procedure
Step 1: 90—110 N·m {9.2—11 kgf·m, 67—81 ft·lbf}
Step 2: 55—65°

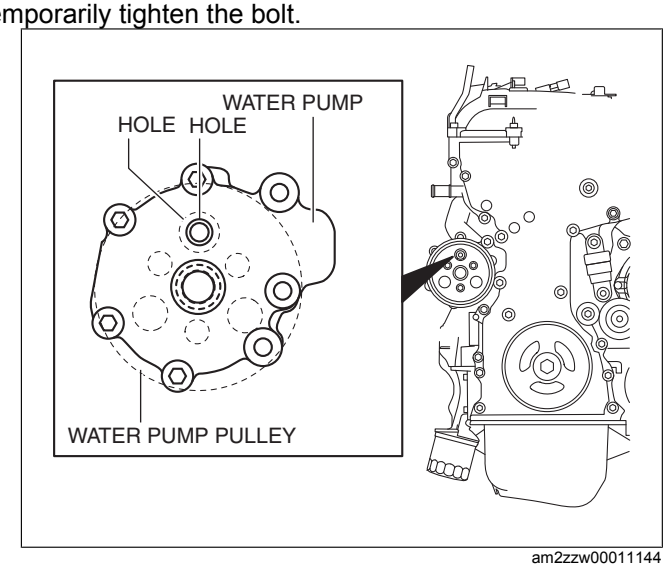


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Water pump pulley installation note

Caution
- **Be careful not to damage the belt groove and surface of the water pump pulley when using tools, otherwise it will cause wear, breakage, abnormal noise of the drive belt (stretch belt), damage to the pulley, and rust.**

1. Install the water pump pulley to the water pump and temporarily tighten the bolt.
2. Align the water pump pulley hole with the water pump hole as shown in the figure.

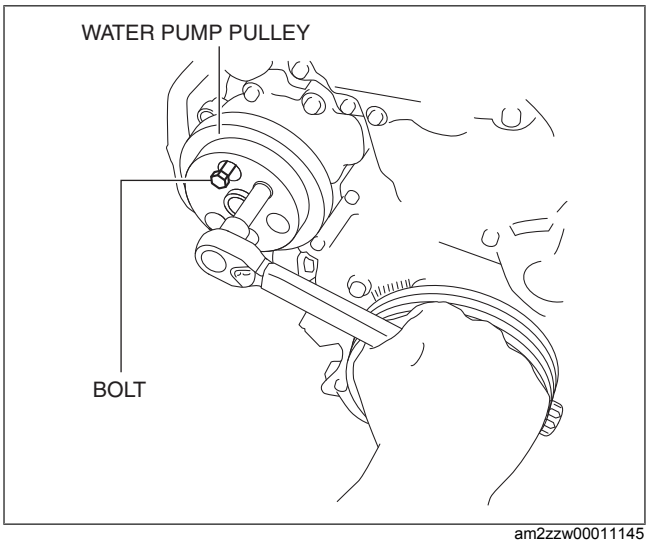


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3. Insert an appropriate bolt (length **70 mm {2.8 in}**) into the water pump hole shown in the figure and lock the water pump pulley against rotation.
4. Completely tighten the water pump pulley bolt to the specified torque.

   Tightening torque
   8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}

5. Remove the bolt used for locking the water pump pulley against rotation.



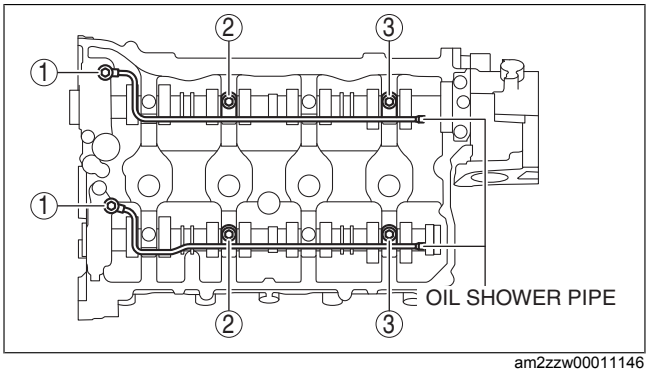
am2zzw00011145

Oil shower pipe installation note

1. Install the oil shower pipe in the order shown in the figure.

Tightening torque

Installation position	Tightening torque
1	9—12 N·m {92—122 kgf·cm, 80—106 in·lbf}
2, 3	8.0—9.0 N·m {82—91 kgf·cm, 71—79 in·lbf}



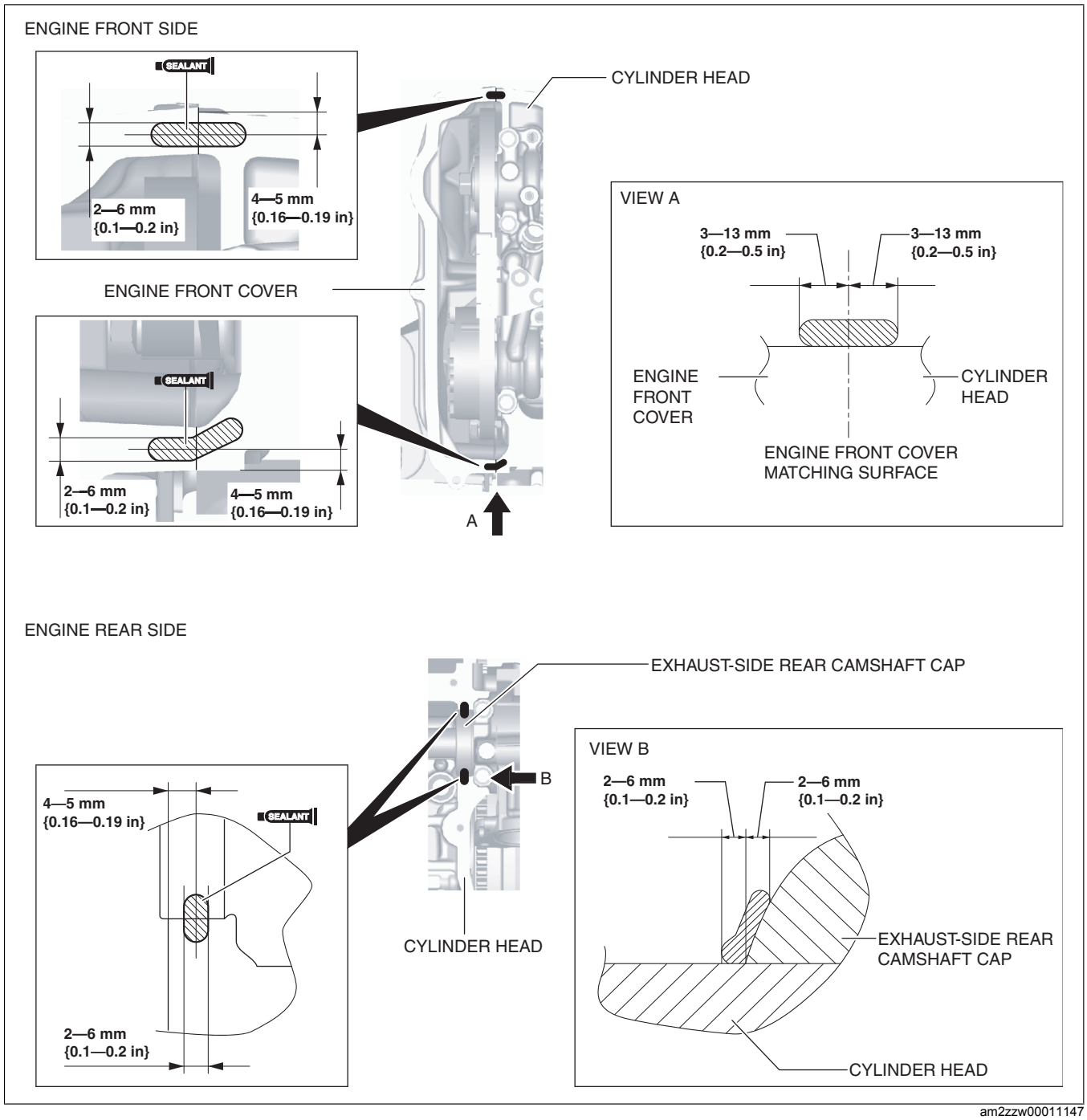
am2zzw00011146

Cylinder head cover installation note

Caution

- **To assure the sealing performance of the cylinder head cover, be careful of the following:**
  - **— Verify that the cylinder head cover gasket is inserted into the cylinder head cover groove and install the cylinder head cover.**
  - **— Completely clean and remove oil, dirt, sealant or other foreign matter from the seal surface.**
- **To prevent silicone sealant from hardening, adhere the cylinder head cover and the cylinder head firmly within 10 min. after applying silicone sealant. After adhering them, tighten the installation bolts immediately.**

1. Insert a new cylinder head cover gasket into the cylinder head cover groove.
2. Apply silicone sealant (TB1217D or equivalent) to the areas shown in the figure.



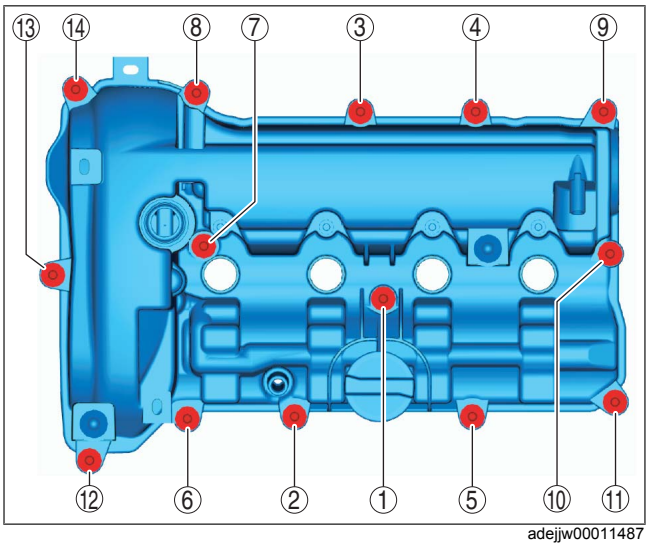
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3. Install the cylinder head cover.
4. Temporarily tighten the cylinder head cover bolts.

5. Tighten the cylinder head cover bolts in the order shown in the figure.

   Tightening torque
   6.0—7.0 N·m {62—71 kgf·cm, 54—61 in·lbf}

6. Measure the tightening torque again and verify that it is **6.0 N·m {61 kgf·cm, 53 in·lbf}** or more.
7. Connect the vacuum hose.
8. Connect the ventilation hose.
9. Return the wiring harness which was set away from the cylinder head cover and the intake manifold to its original position.
10. Connect the OCV connector.

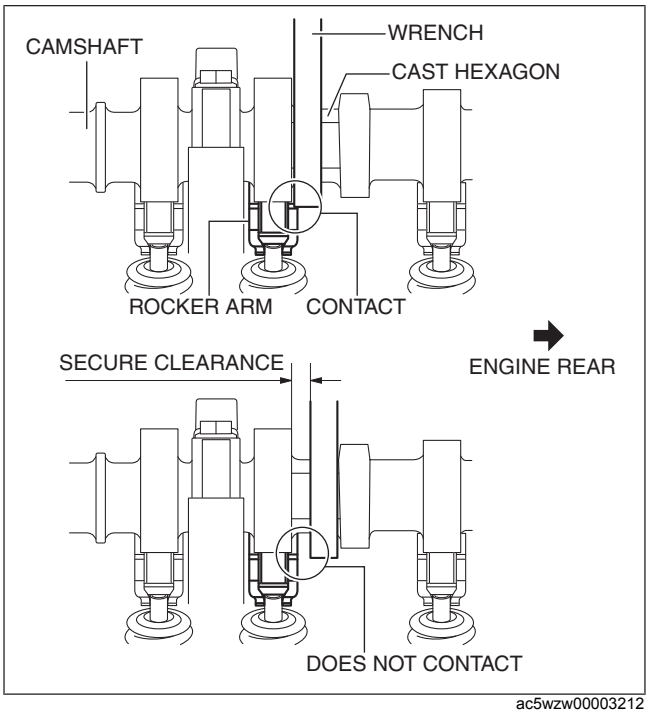


adejjw00011487

SKYACTIV-G 1.5 (with 4-2-1 exhaust system)

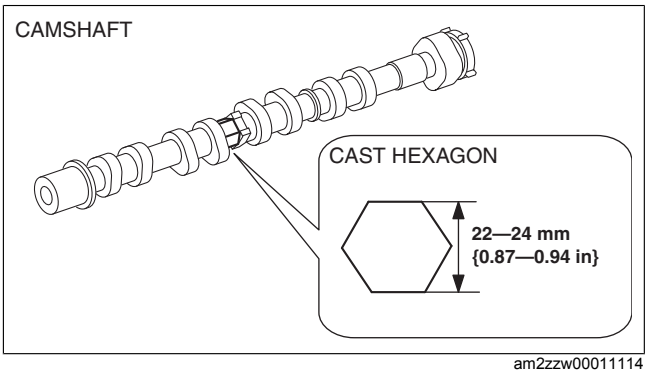
Warning
- **A hot engine can cause severe burns. Turn off the engine and wait until it is cool before servicing.**

Caution
- **If the camshaft is rotated with the timing chain removed and the piston at the top dead center position, the valve may contact the piston and the engine could be damaged. When rotating the camshaft with the timing chain removed, rotate it after lowering the piston from the top dead center position.**
- **When rotating the camshaft using a wrench on the cast hexagon, the wrench may contact the rocker arm and damage the rocker arm. To prevent damage to the rocker arm when holding the camshaft on the cast hexagon, use a wrench on the rear side of the engine as shown in the figure to secure a clearance between the cam.**



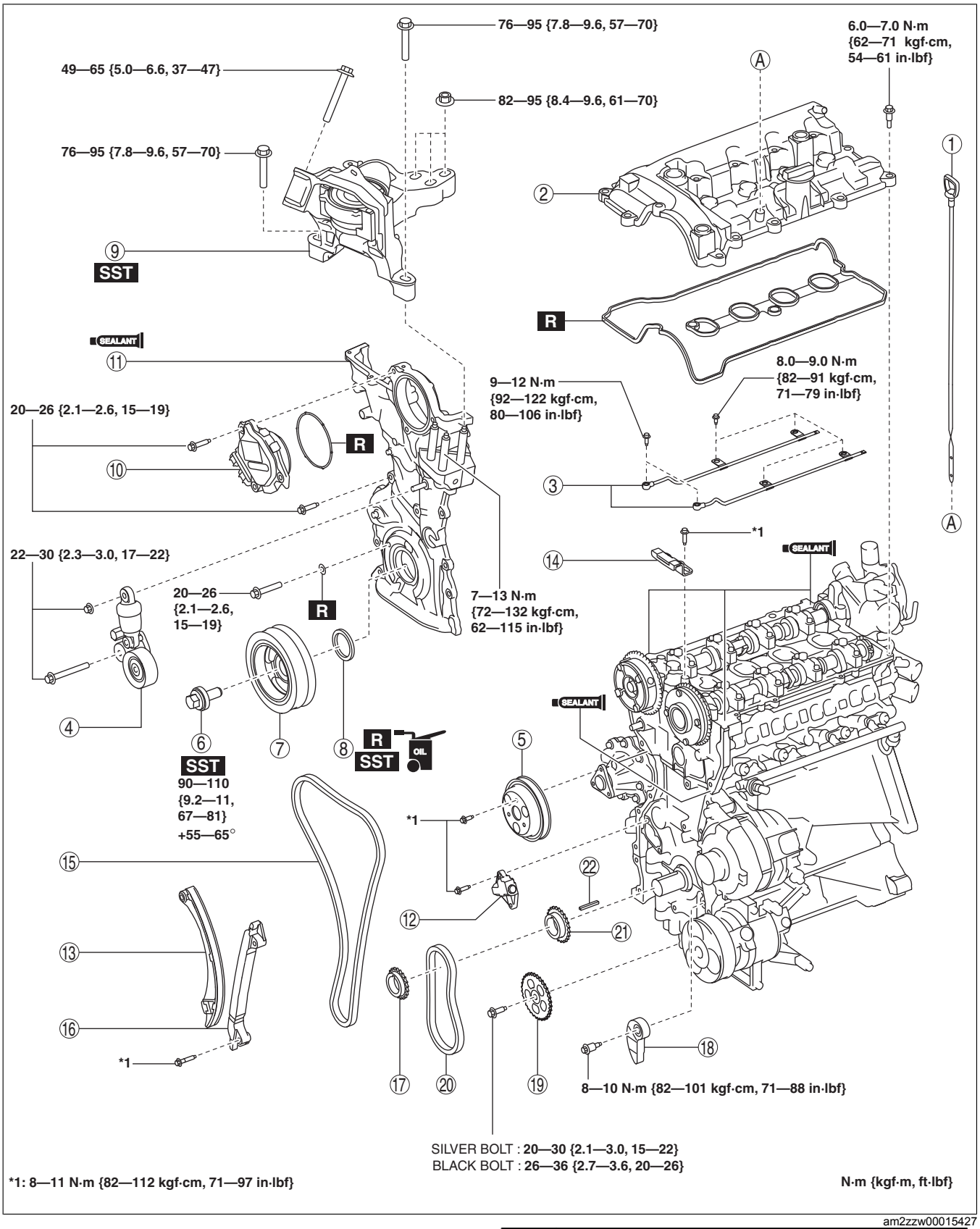
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Note
- Width at the cast hexagon of the camshaft is **22—24 mm {0.87—0.94 in}**.



am2zzw00011114

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
2. Remove the plug hole plate. (See PLUG HOLE PLATE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
3. Remove the ignition coil/ion sensors. (See IGNITION COIL/ION SENSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
4. Remove the front under cover No.2. (See FRONT UNDER COVER No.2 REMOVAL/INSTALLATION.)
5. Remove the splash shield (RH). (See SPLASH SHIELD REMOVAL/INSTALLATION.)
6. Remove the drive belt. (See DRIVE BELT REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
7. Drain the engine oil. (See ENGINE OIL REPLACEMENT [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
8. Remove the oil pan. (See OIL PAN REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
9. Remove in the order indicated in the table.
10. Install in the reverse order of removal.
11. Refill with the specified type and amount of the engine oil. (See ENGINE OIL REPLACEMENT [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
12. Start the engine and inspect the following:
    - Leakage of engine oil.
    - Runout and contact of pulley and belt.
    - Ignition timing, idle speed and idle mixture. (See ENGINE TUNE-UP [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)



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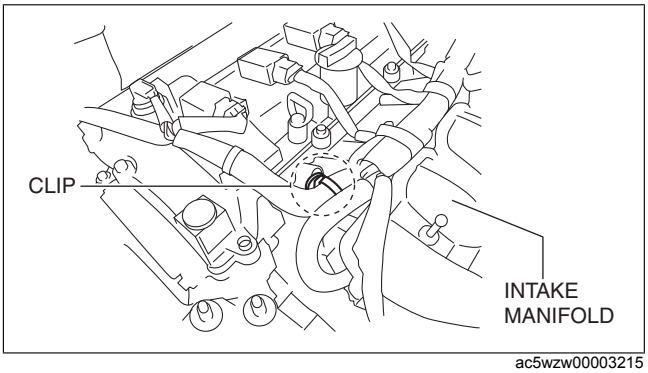
1	Dipstick
2	Cylinder head cover (See Cylinder head cover removal note.) (See Cylinder head cover installation note.)
3	Oil shower pipe (See Oil shower pipe installation note.)

4	Drive belt auto tensioner (See DRIVE BELT AUTO TENSIONER REMOVAL/ INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
5	Water pump pulley (See Water pump pulley removal note.) (See Water pump pulley installation note.)

6	Crankshaft pulley lock bolt (See Crankshaft pulley lock bolt removal note.) (See Crankshaft pulley lock bolt installation note.)
7	Crankshaft pulley
8	Front oil seal (See FRONT OIL SEAL REPLACEMENT [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
9	No.3 engine mount (See No.3 engine mount removal note.) (See No.3 engine mount installation note.)
10	Electric variable valve timing motor/driver (See Electric variable valve timing motor/driver installation note.)
11	Engine front cover (See Engine front cover removal note.) (See Engine front cover installation note.)
12	Chain tensioner (See Timing chain removal note.) (See Timing chain installation note.)
13	Tensioner arm (See Timing chain removal note.) (See Timing chain installation note.)
14	Chain guide (No.1) (See Timing chain removal note.) (See Timing chain installation note.)
15	Timing chain (See Timing chain removal note.) (See Timing chain installation note.)
16	Chain guide (No.2)
17	Crankshaft sprocket
18	Oil pump chain tensioner
19	Oil pump driven sprocket (See Oil pump driven sprocket removal note.) (See Oil pump driven sprocket installation note.)
20	Oil pump chain
21	Oil pump drive sprocket
22	Key

Cylinder head cover removal note

1. Disconnect the OCV connector.
2. Remove the wiring harness clips secured to the cylinder head cover.
3. Remove the wiring harness clip secured to the intake manifold as shown in the figure, and set the wiring harness aside.
4. Disconnect the ventilation hose.
5. Disconnect the brake vacuum hose from the intake manifold and set it aside.
6. Remove the cylinder head cover.



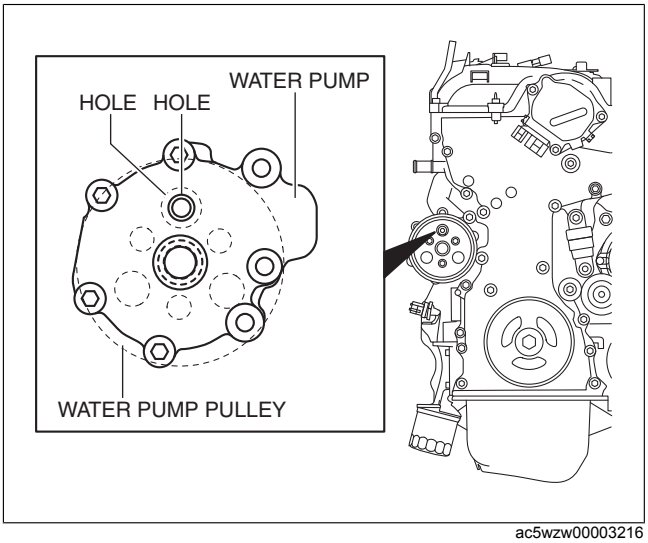
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Water pump pulley removal note

Caution

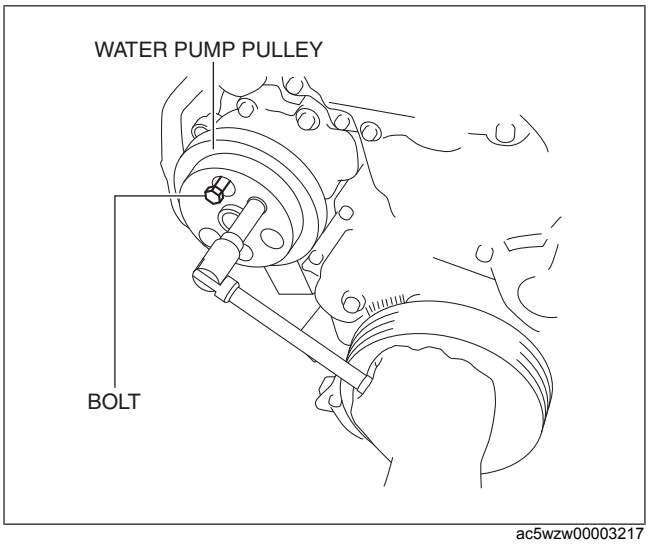
- **Be careful not to damage the belt groove and surface of the water pump pulley when using tools, otherwise it will cause wear, breakage, abnormal noise of the drive belt (stretch belt), damage to the pulley, and rust.**

1. Align the water pump pulley hole with the water pump hole as shown in the figure.



ac5wzw00003216

2. Insert an appropriate bolt (length **approx. 70 mm {2.8 in}**) into the water pump hole as shown in the figure, and lock the water pump pulley against rotation.
3. Remove the water pump pulley.
4. Remove the bolt used for locking the water pump pulley against rotation.

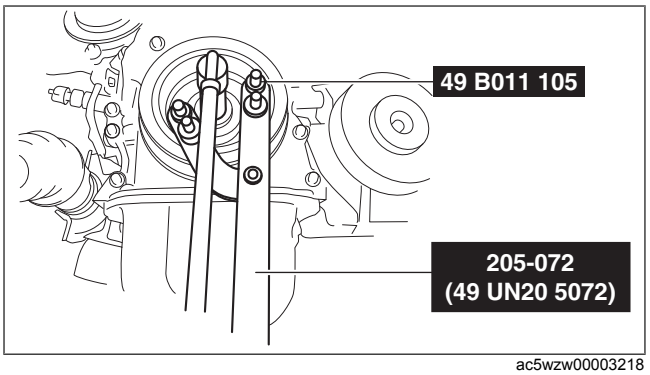


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Crankshaft pulley lock bolt removal note

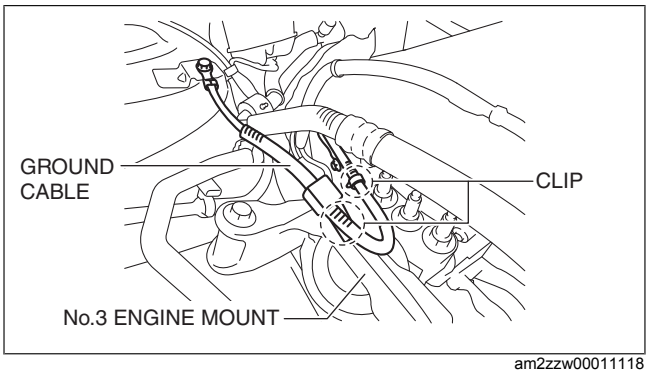
Caution
- **To prevent damage to the crankshaft pulley, protect the crankshaft pulley with a clean cloth so that the bar part (205-072) of the SST does not contact the crankshaft pulley.**
- **To prevent damage to the engine front cover, be careful not to allow the tab part (49 B011 105) of the SST to get caught between the crankshaft pulley and the engine front cover.**

1. Install the **SST** to the crankshaft pulley, and lock the crankshaft against rotation.
2. Remove the crankshaft pulley lock bolt.



ac5wzw00003218

No.3 engine mount removal note
1. Remove the clips shown in the figure and set the ground cable aside.

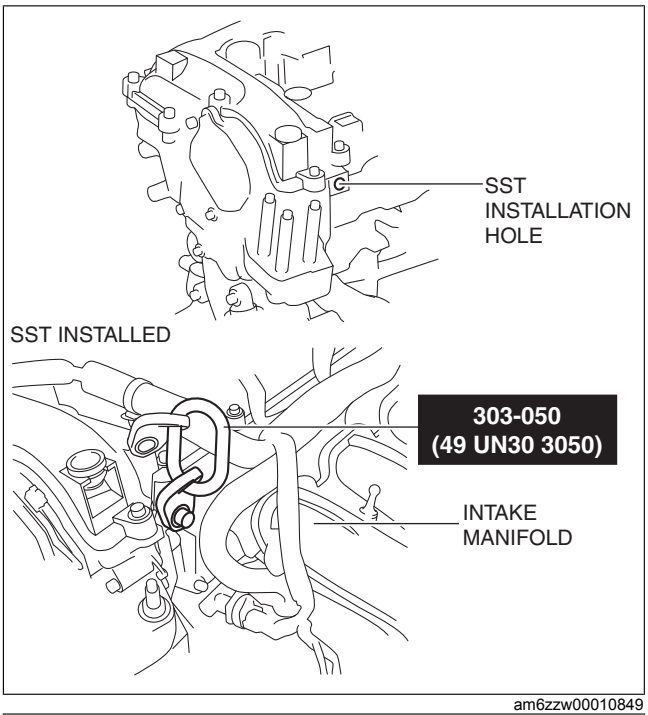


am2zzw00011118

2. Install the **SST** and the following bolt to the position shown in the figure.

   Bolt:
   part number 99794 1025 or an M10 × 1.25, length 25 mm {0.98 in}
   Tightening torque
   38—51 N·m {3.9—5.2 kgf·m, 29—37 ft·lbf}

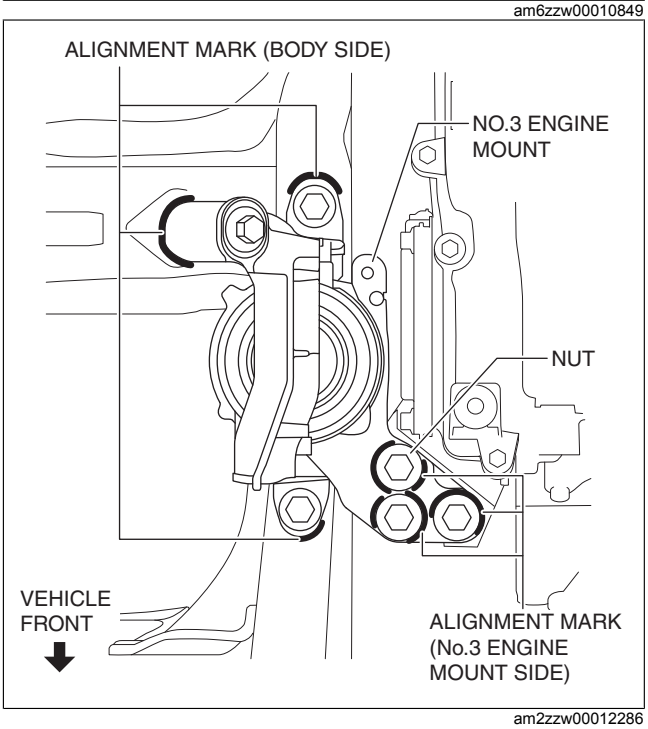
   Caution
   - **Slots have been adopted for the No.3 engine mount installation holes. If the No.3 engine mount is deviated from the original position when installing the No.3 engine mount, engine noise or vibration could increase. Before removing the No.3 engine mount, place alignment marks on the No.3 engine mount and body so that they can be assembled to the same positions as before removal.**



am6zzw00010849

3. Place alignment marks on the locations shown in the figure so that they can be assembled to the same positions as before removal.

   Note
   - Paint so that the No.3 engine mount is framed on the body side and the outline of the nut is framed on the No.3 engine mount side.



am2zzw00012286

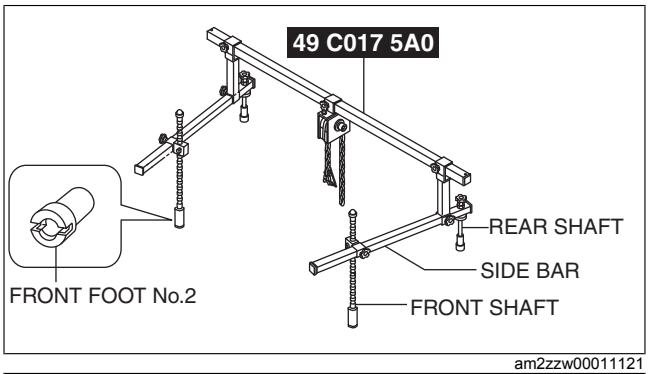
4. Install the **SST** using the following procedures.

   Caution
   - **Refer to the SST instruction manual for the basic handing procedure.**

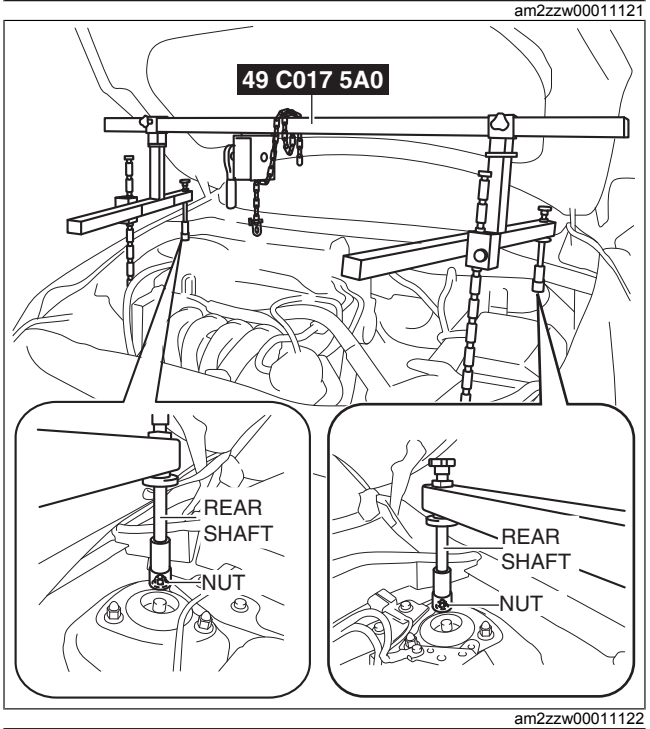
Note

- Install front foot No. 2 to the left and right front shafts of the **SST**.

(1) Set the rear shafts of the **SST (49 C017 5A0)** on the left/right front shock absorber nuts (top) as shown in the figure.

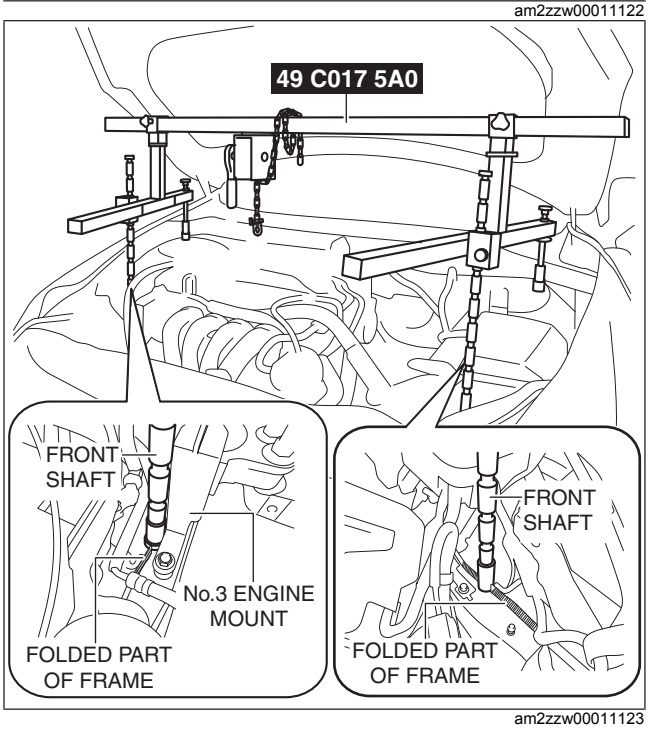


am2zzw00011121



am2zzw00011122

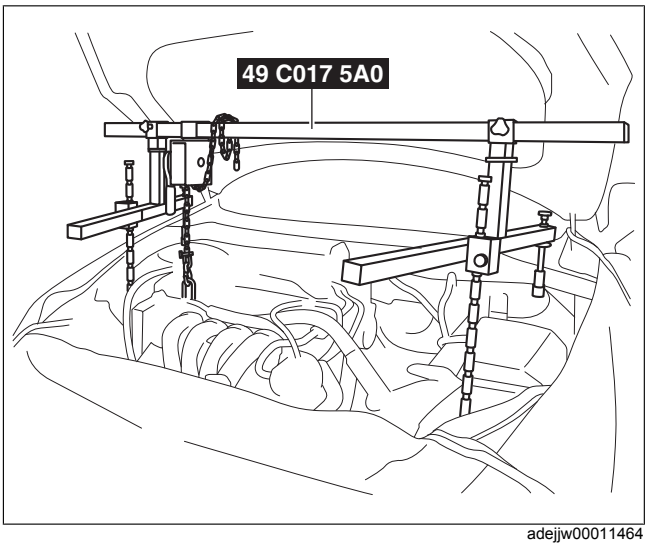
(2) Set the front shafts of the **SST (49 C017 5A0)** on the bent up part of the frame as shown in the figure.



am2zzw00011123

(3) Adjust the height of the left and right side bars of the **SST** so that they are leveled, then tighten each part.
(4) Apply tension to the chain to hold the engine.

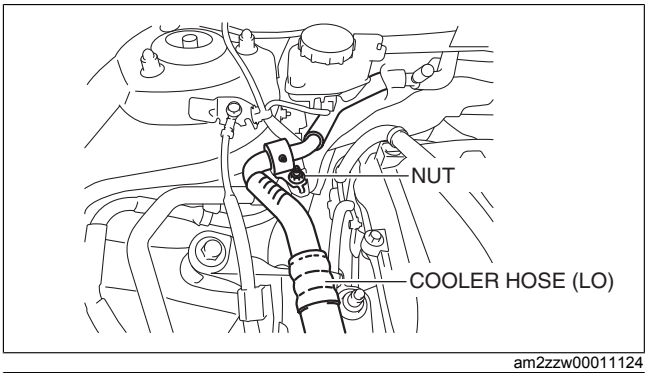
5. Remove the No.3 engine mount.



adejjw00011464

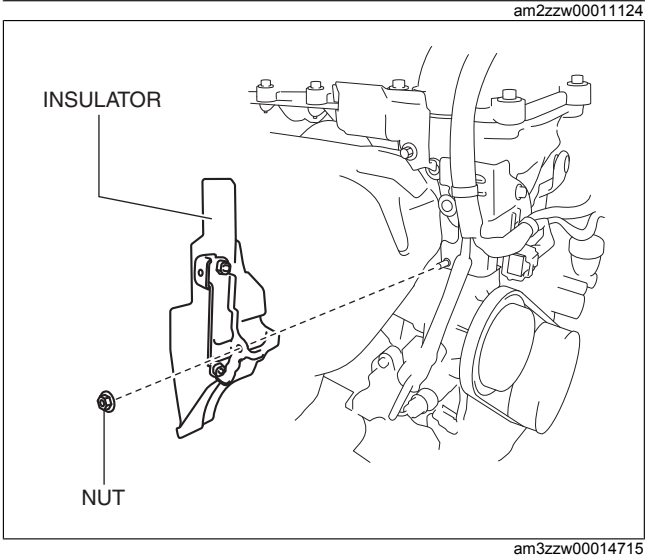
Engine front cover removal note

1. Remove the nut shown in the figure and set the cooler hose (LO) aside.



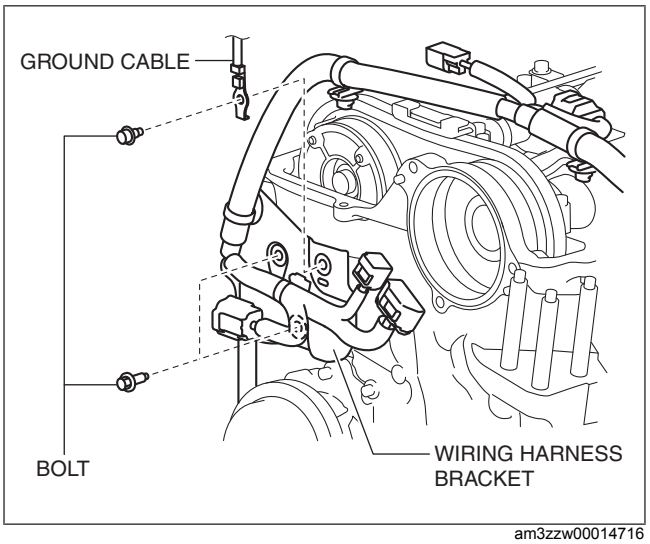
am2zzw00011124

2. Remove the insulator shown in the figure.



am3zzw00014715

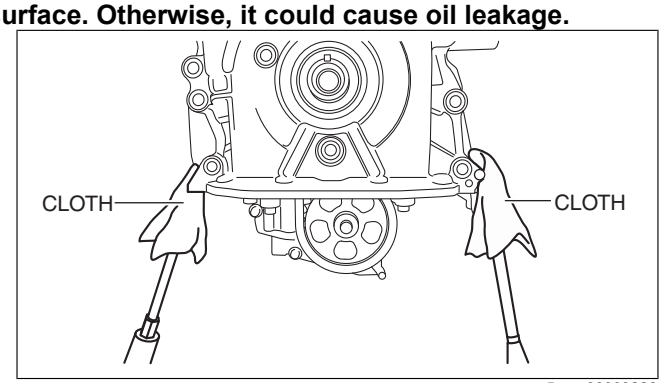
3. Set the ground cable and wiring harness bracket shown in the figure aside.
4. Remove the engine front cover installation bolts.



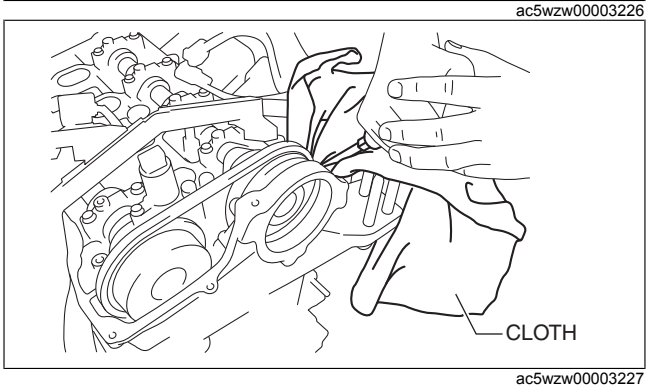
am3zzw00014716

5. Using a screwdriver wrapped in a cloth, peel the sealant away a little at a time, and remove the engine front cover.

Caution
- **Do not apply excessive force to the screwdriver. Otherwise, the engine front cover could be damaged.**
- **Be careful not to scratch or damage the seal surface. Otherwise, it could cause oil leakage.**

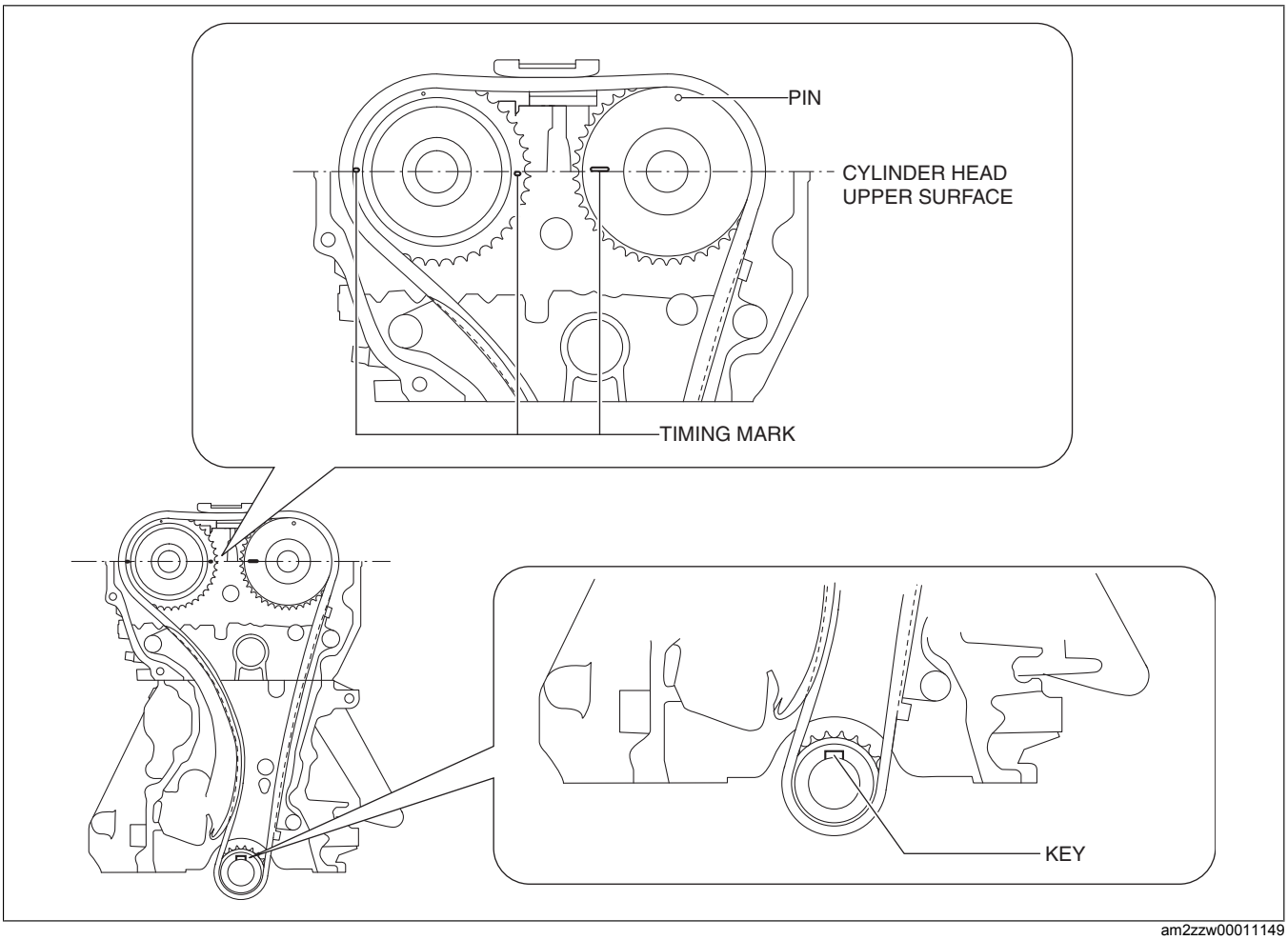


ac5wzw00003226



ac5wzw00003227

Timing chain removal note
1. Rotate the crankshaft clockwise to align the timing marks and the key position as shown in the figure, and set cylinder No.1 at top dead center (TDC).

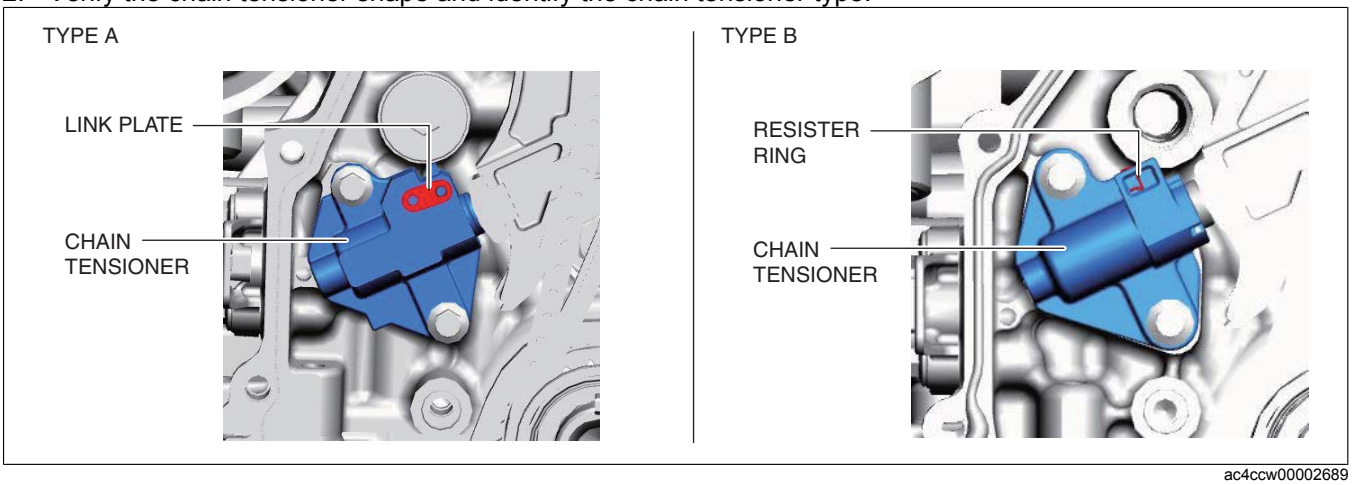


am2zzw00011149

Note

- The timing mark is not completely parallel with the upper surface of the cylinder head.

2. Verify the chain tensioner shape and identify the chain tensioner type.

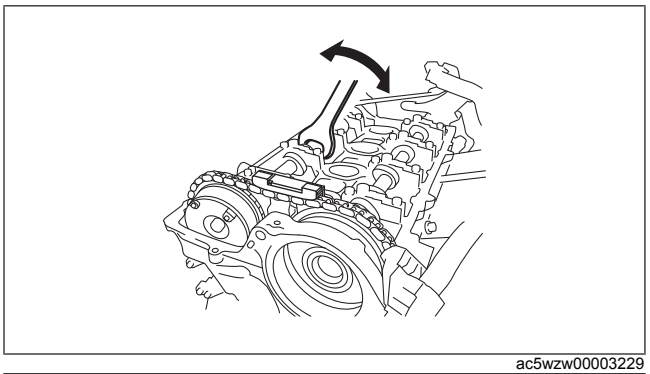


ac4ccw00002689

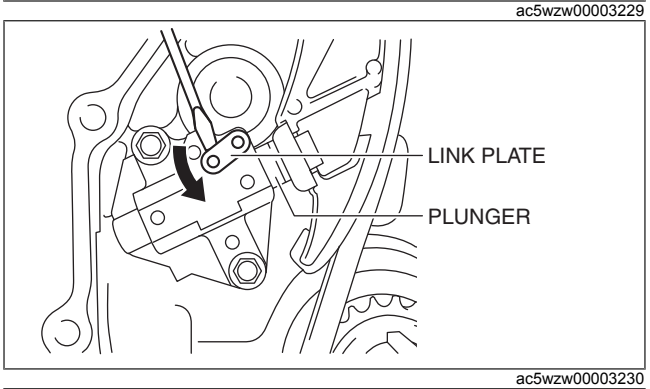
Note

- Verifying the chain tensioner type is necessary because the following procedures may differ according to the type of tensioner.

3. While moving the exhaust camshaft back and forth in the direction of the arrow using a wrench on the cast hexagon, press down the link plate of the timing chain tensioner using a precision screwdriver and release the plunger lock. (Chain tensioner (type A))

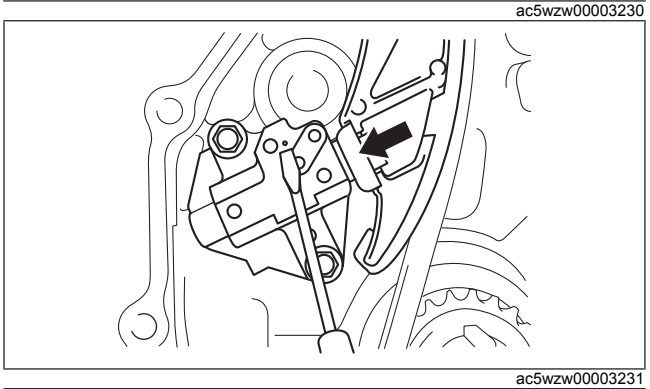

ac5wzw00003229

Note
- When moving the exhaust camshaft back and forth, the timing chain pushes the plunger in the chain tensioner making it easier to operate the link plate.

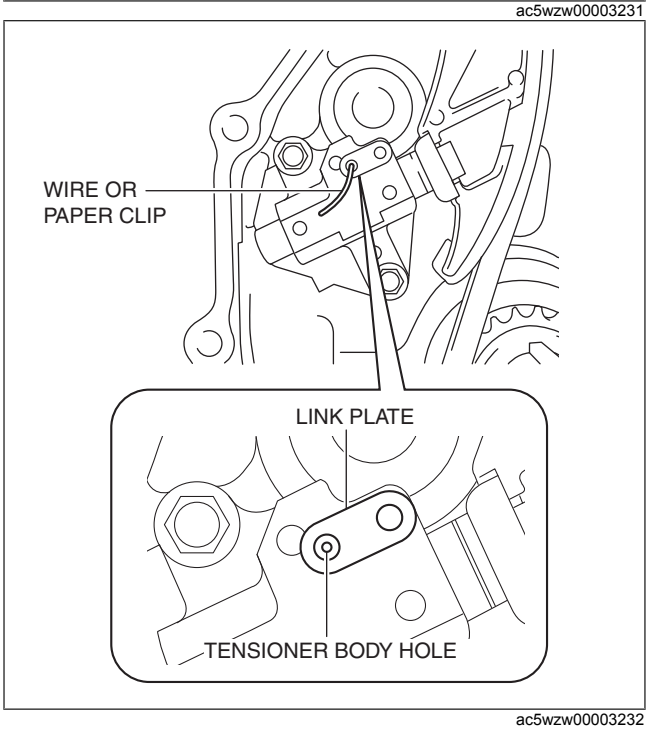


ac5wzw00003230

4. Push back the plunger slowly in the direction shown in the figure with the link plate still pushed down. (Chain tensioner (type A))
5. Remove the screwdriver from the link plate with the plunger still pushed down. (Chain tensioner (type A))
6. Release the force slightly from the plunger, and move it back and forth **2—3 mm {0.08—0.11 in}**. (Chain tensioner (type A))


ac5wzw00003231

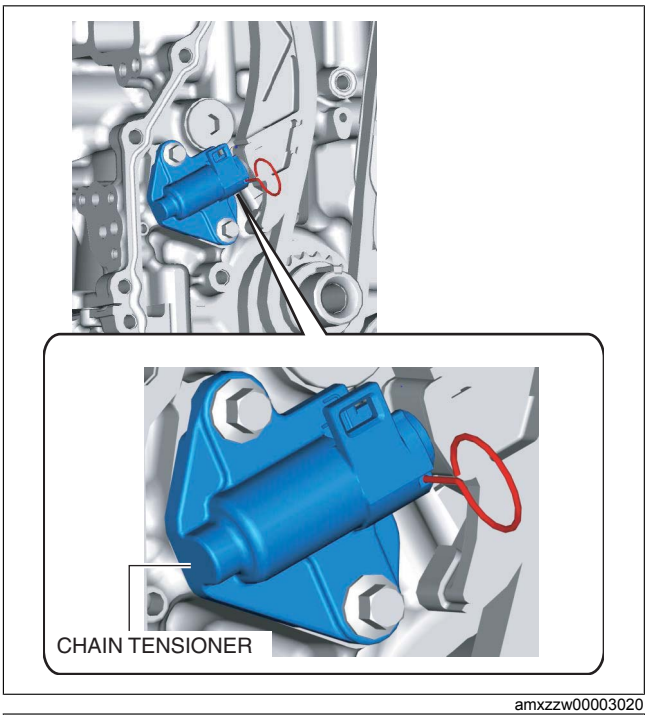
7. Insert a wire with an **approx. diameter of 1.5 mm {0.059 in}** or a paper clip where the link plate hole and the tensioner body hole overlap to secure the link plate and lock the plunger. (Chain tensioner (type A))



ac5wzw00003232

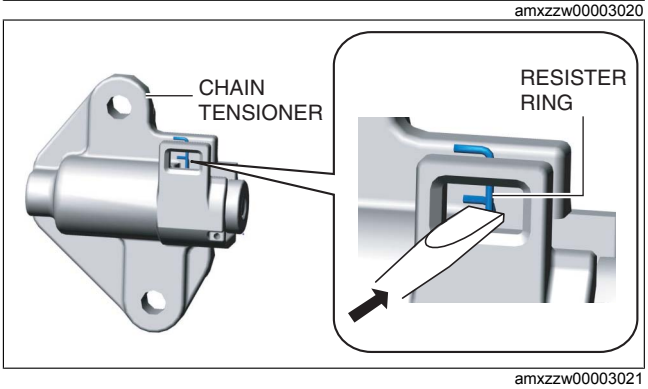
8. Loosen the chain tensioner using the following procedure: (Chain tensioner (type B))

(1) Insert a metal rod (diameter **approx. 1.4 mm {0.055 in}**, length **approx. 55 mm {2.2 in}**) into the hole in the body of the chain tensioner.



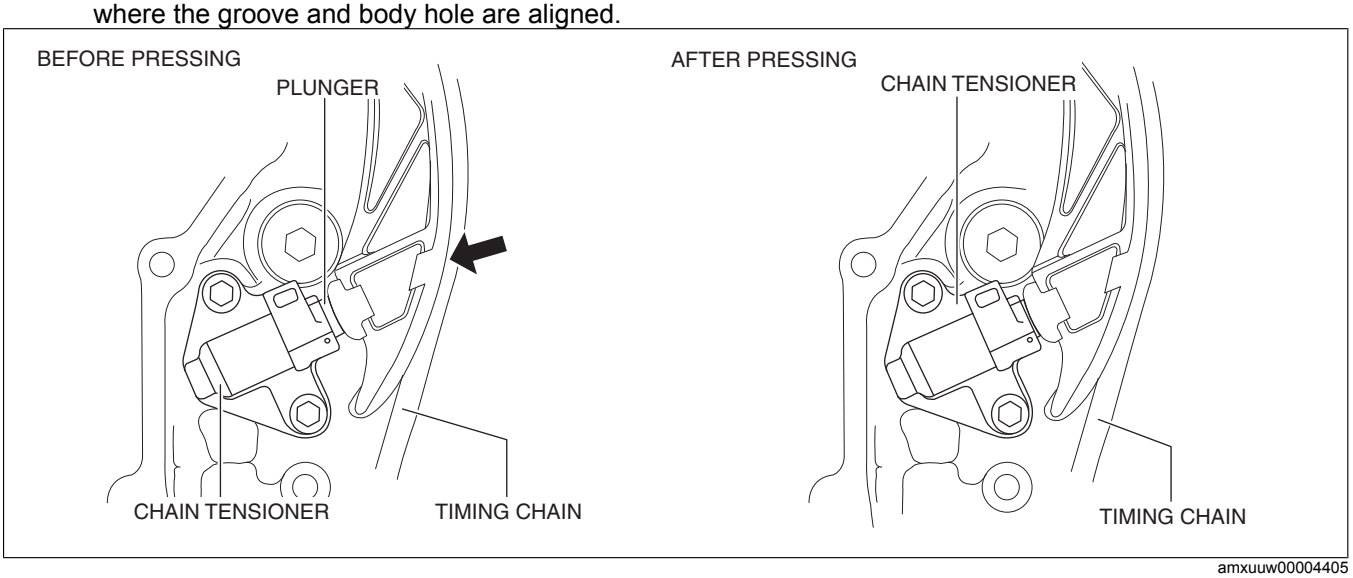
amxzzw00003020

(2) Press the resister ring and unlock the plunger.



amxzzw00003021

(3) Press the timing chain in the direction of the arrow and press in the chain tensioner plunger to the position where the groove and body hole are aligned.



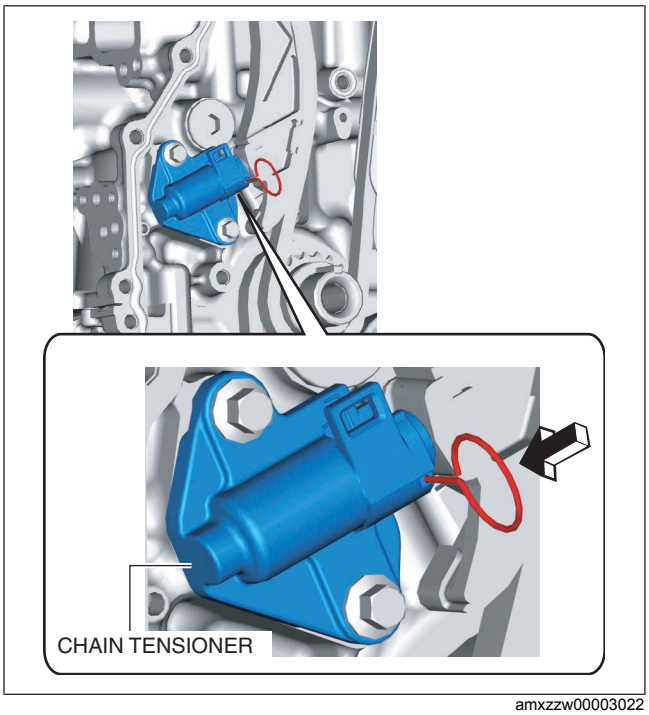
amxuuw00004405

(4) With the plunger pressed in, further insert the metal rod set in (1) above.

Note
- The rod secures the plunger and releases the tension.

(5) Loosen the power of the hand pressing the plunger and verify that the pressed-in rod does not move.

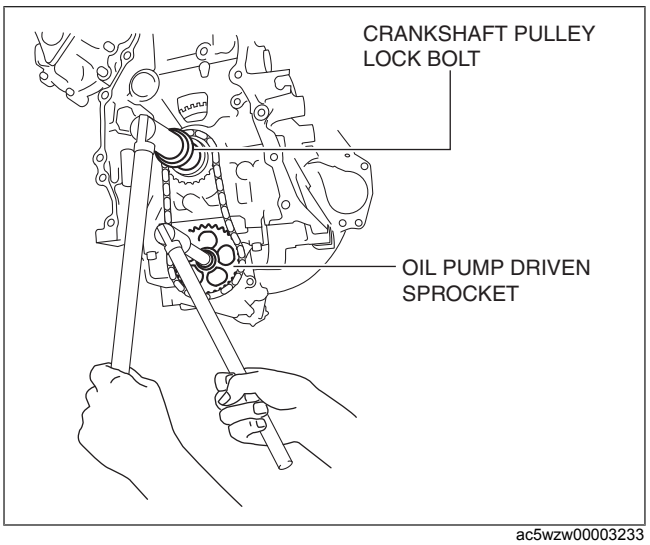
9. Remove the chain tensioner, tensioner arm, and the chain guide (No.1).
10. Remove the timing chain.



amxzzw00003022

Oil pump driven sprocket removal note

1. Temporarily assemble the crankshaft pulley lock bolt, and lock the oil pump against rotation as shown in the figure.
2. Remove the oil pump driven sprocket.
3. Remove the temporarily assembled crankshaft pulley lock bolt.



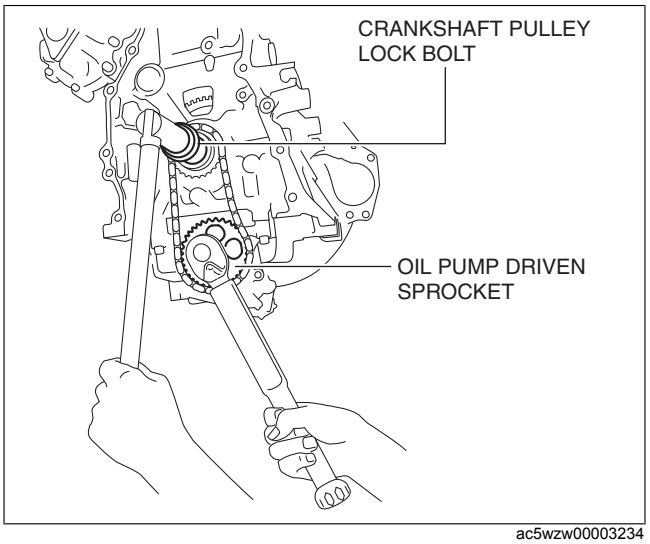
ac5wzw00003233

Oil pump driven sprocket installation note

1. Temporarily assemble the crankshaft pulley lock bolt, and lock the oil pump against rotation as shown in the figure.
2. Install the oil pump driven sprocket.

Tightening torque
Silver bolt: 20—30 N·m {2.1—3.0 kgf·m, 15—22 ft·lbf}
Black bolt: 26—36 N·m {2.7—3.6 kgf·m, 20—26 ft·lbf}

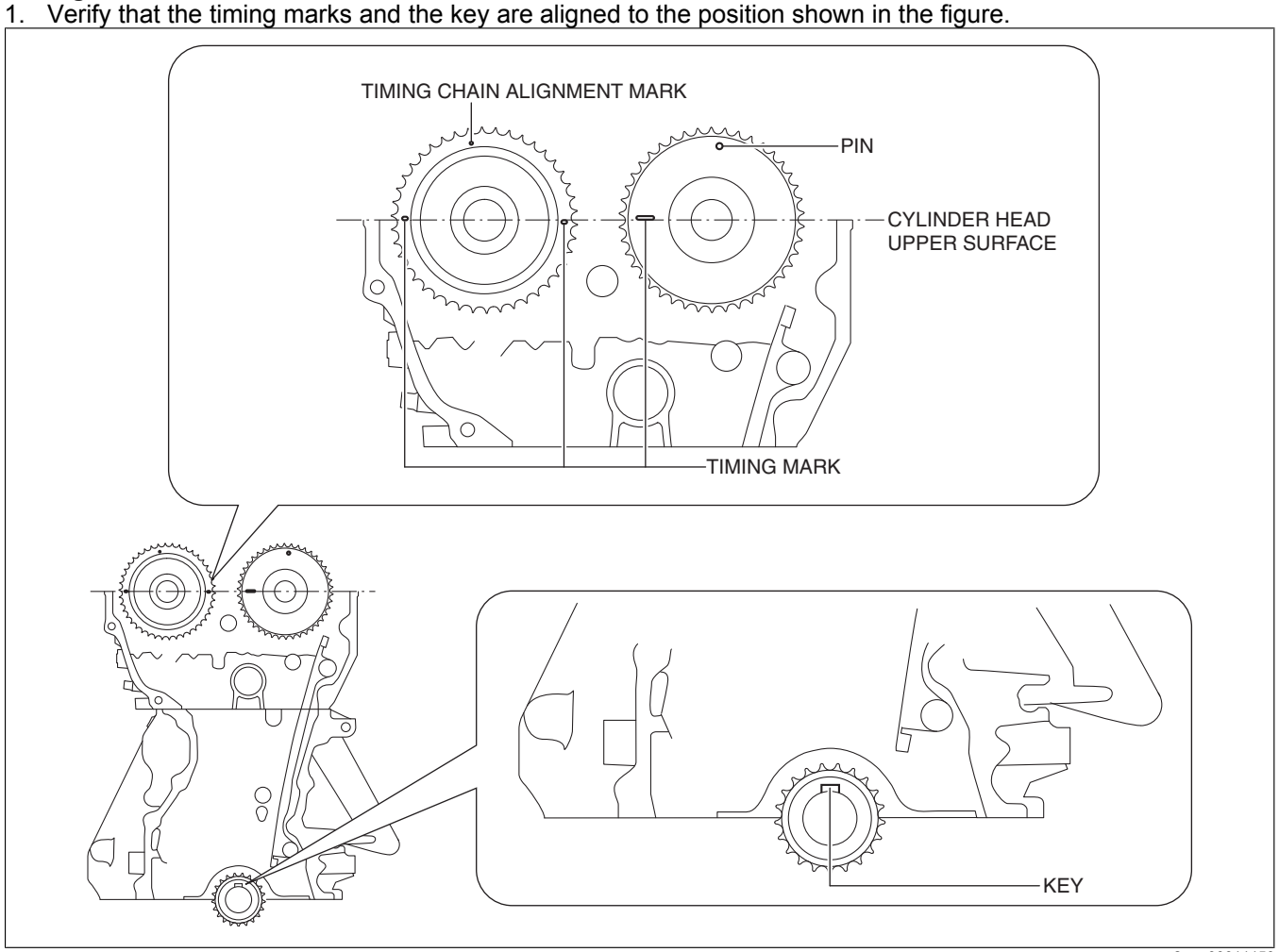
3. Remove the temporarily assembled crankshaft pulley lock bolt.



ac5wzw00003234

Timing chain installation note

1. Verify that the timing marks and the key are aligned to the position shown in the figure.

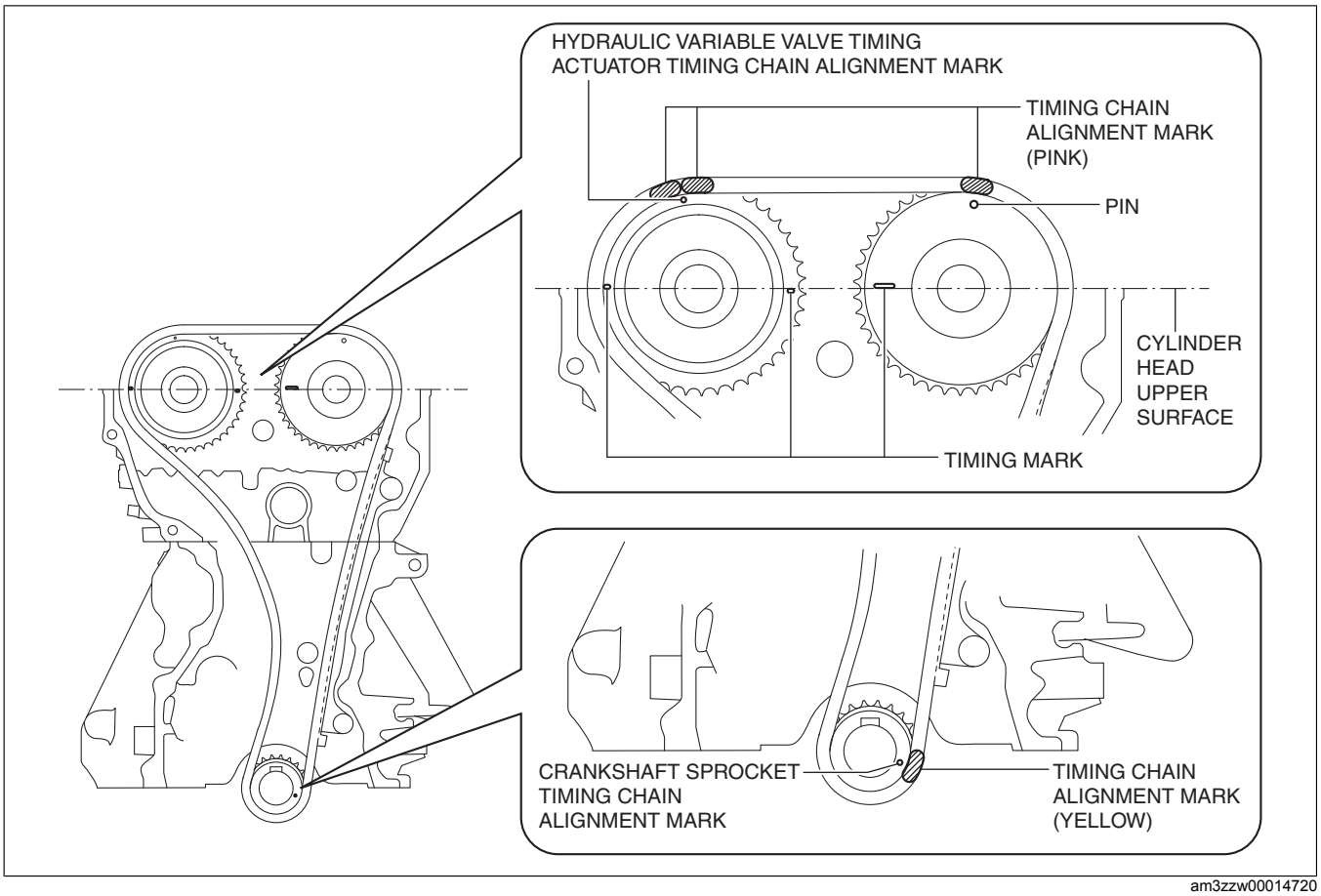


am2zzw00011150

Note

- The timing mark is not completely parallel with the upper surface of the cylinder head.
- If they are not in the position shown in the figure, rotate the camshaft and crankshaft to set the cylinder No. 1 top dead center (TDC).

2. Install the timing chain while aligning the marks on the timing chain as shown in the figure.



am3zzw00014720

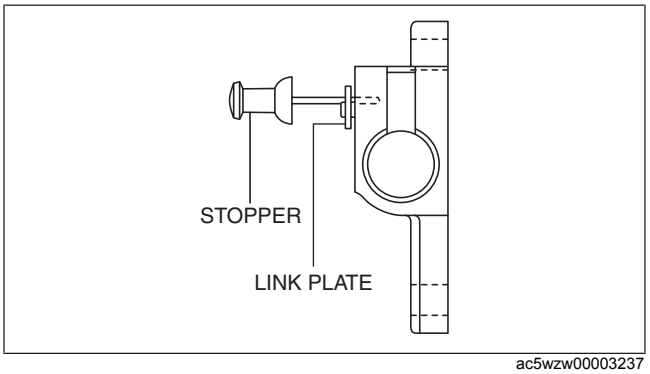
3. Install the chain guide (No.1).

   Tightening torque
   8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}

4. Install the tensioner arm.
5. Install the chain tensioner.

   Tightening torque
   8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}

6. After installing the chain tensioner, remove the installed wire or paper clip, and then apply tension to the timing chain. (Chain tensioner (type A))
   - If a new chain tensioner is used, remove the installed stopper.

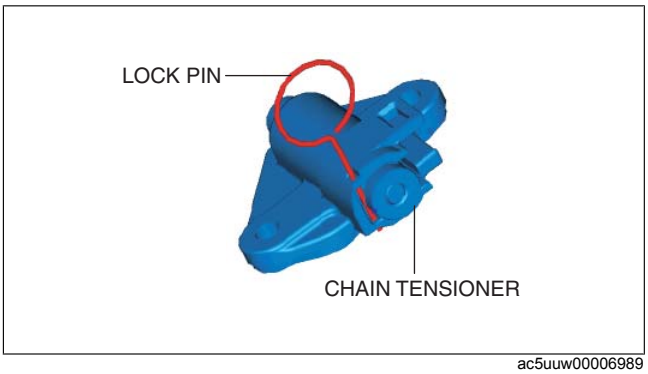


ac5wzw00003237

7. After installing the timing chain tensioner, remove the installed rod, and then apply tension to the timing chain. (Chain tensioner (type B))

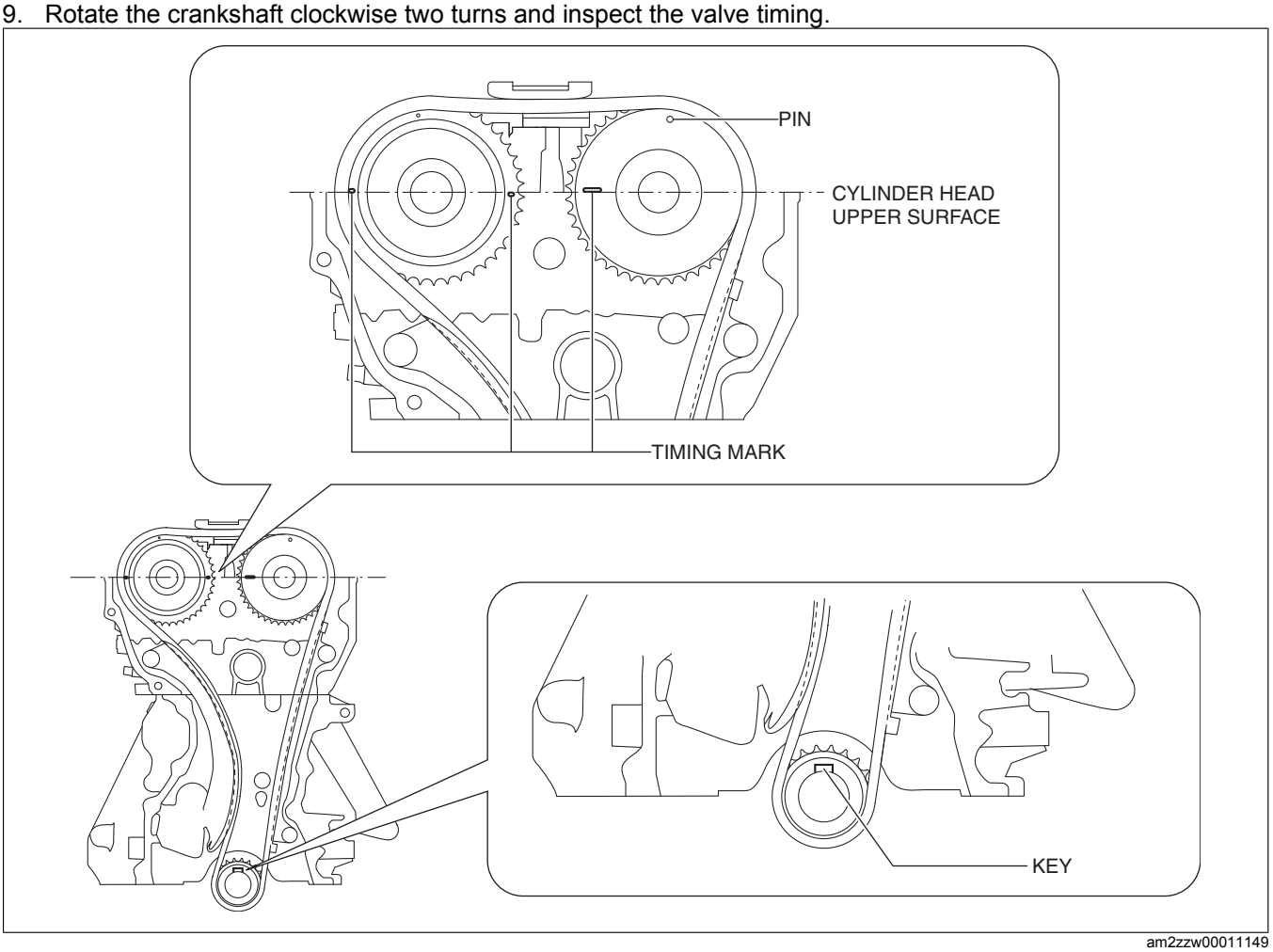
- If a new chain tensioner is used, remove the installed lock pin.

8. Verify that there is no looseness in the timing chain, and re-verify that each sprocket is in the specified location.



ac5uuw00006989

9. Rotate the crankshaft clockwise two turns and inspect the valve timing.



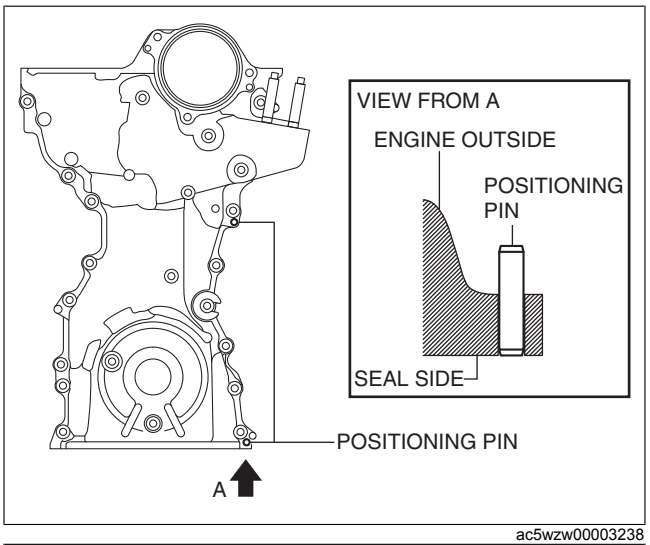
am2zzw00011149

Note

- The timing mark is not completely parallel with the upper surface of the cylinder head.

Engine front cover installation note

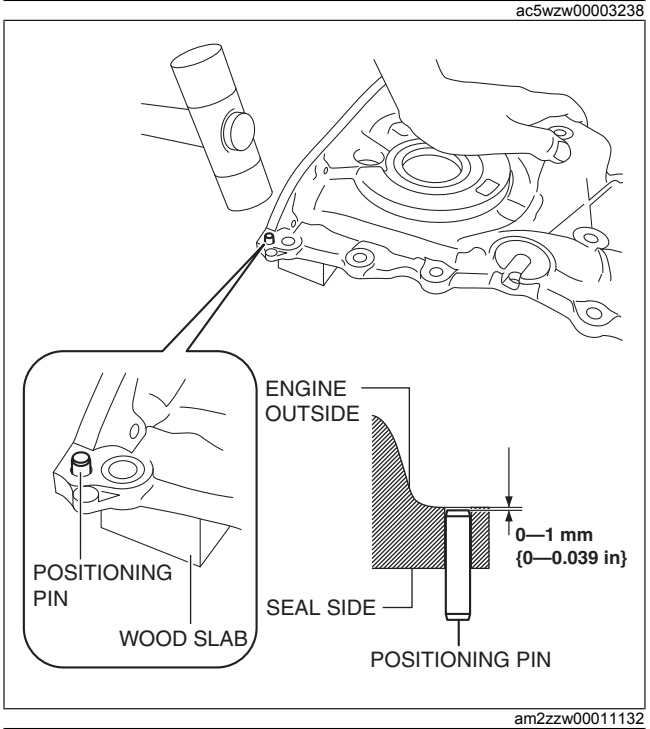
Note
- For a new engine front cover, the positioning pins in the two locations shown in the figure project to the outside of the engine.



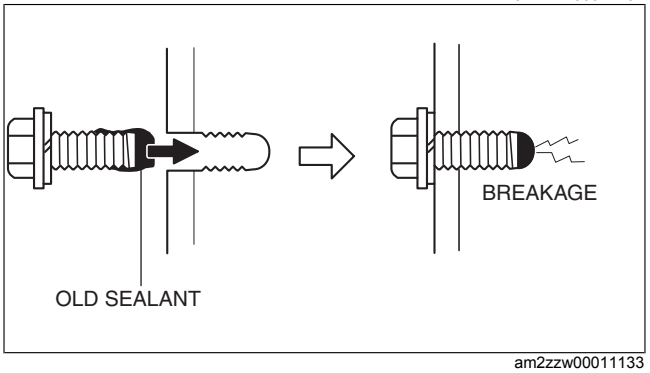
ac5wzw00003238

1. If the engine front cover is newly replaced, tap the positioning pins in the **two locations** to the seal surface side.

Caution
- **If a bolt with sealant adhering to it is used, it could result in cracks in the cylinder head and cylinder block.**



am2zzw00011132



am2zzw00011133

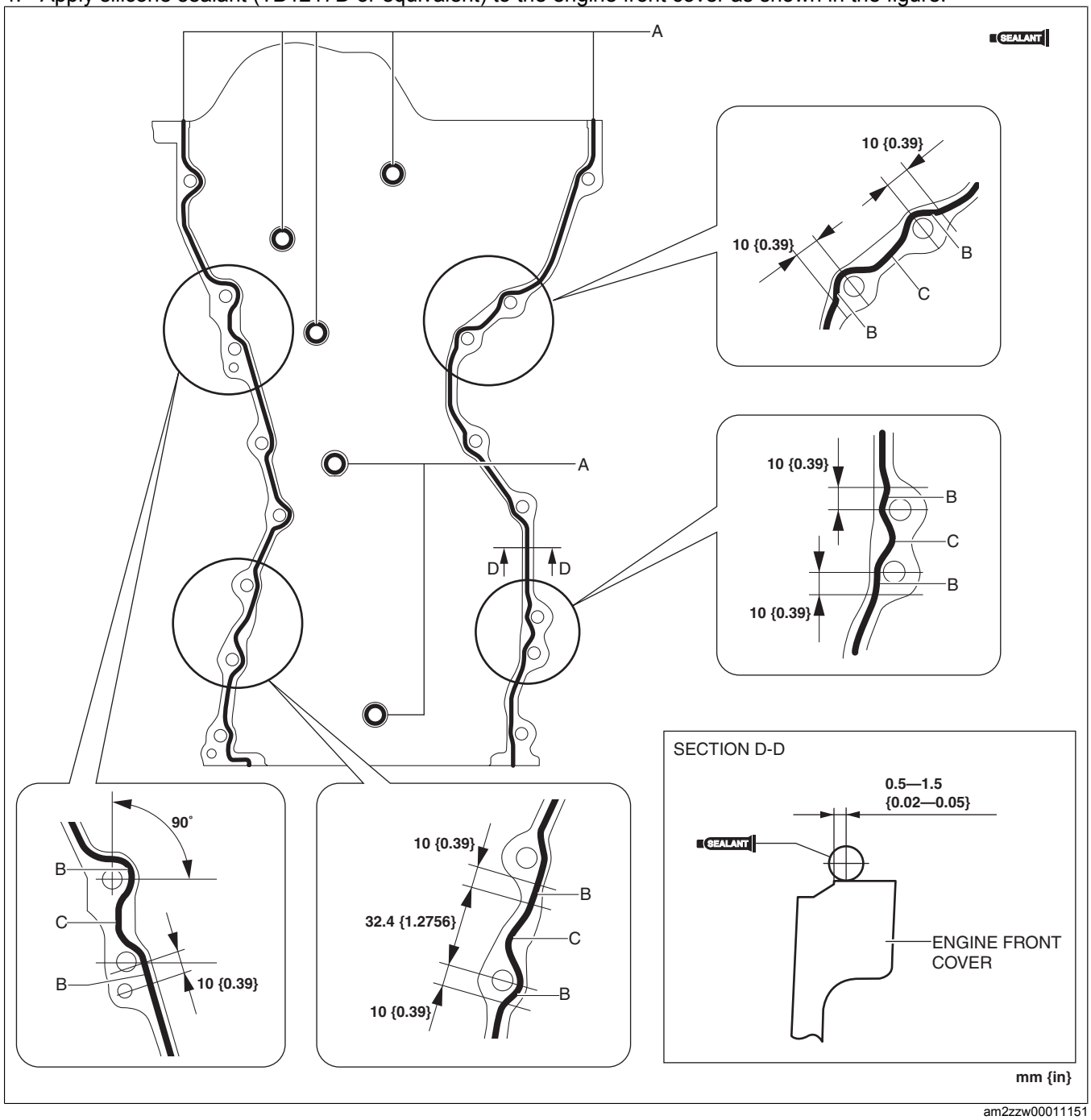
2. When reusing an engine front cover installation bolts, remove sealant adhering to the bolts.

Caution
- **If oil, dirt and sealant remains on the sealant application area, the silicone sealant will not seal which will cause oil leakage.**

3. Completely clean and remove oil, dirt, sealant or other foreign matter that may be adhering to the engine front cover, cylinder head, and cylinder block.

Caution
- **Apply the silicon sealant in a single, unbroken line.**
- **To prevent silicone sealant from hardening, adhere the engine front cover and the cylinder block firmly within 10 min. after applying silicone sealant. After adhering them, tighten the installation bolts immediately.**

4. Apply silicone sealant (TB1217D or equivalent) to the engine front cover as shown in the figure.

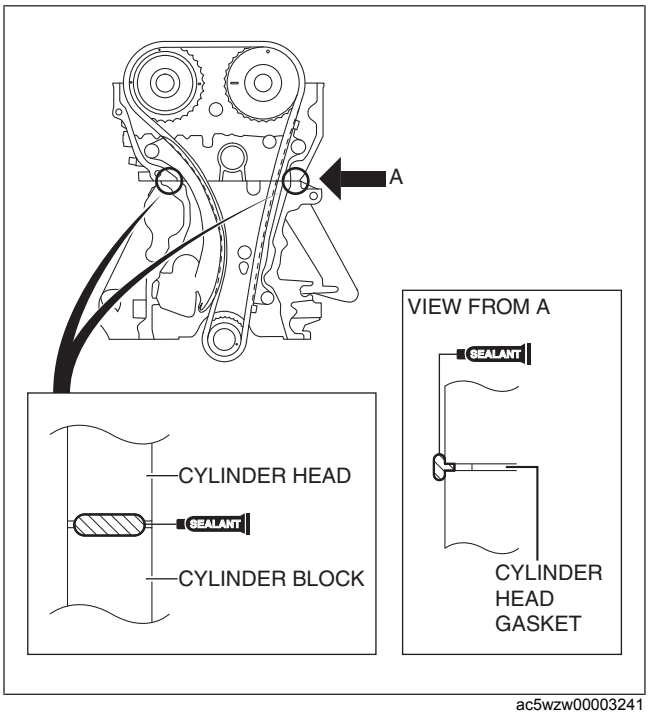


am2zzw00011151

Bead thickness
A: 2—6 mm {0.1—0.2 in}
B: 4—6 mm {0.16—0.23 in}
C: 4—8 mm {0.2—0.3 in}

5. Apply silicone sealant (TB1217D or equivalent) to the areas shown in the figure.

Caution
- **Apply the silicone sealant so that it goes into the cylinder head gasket.**



ac5wzw00003241

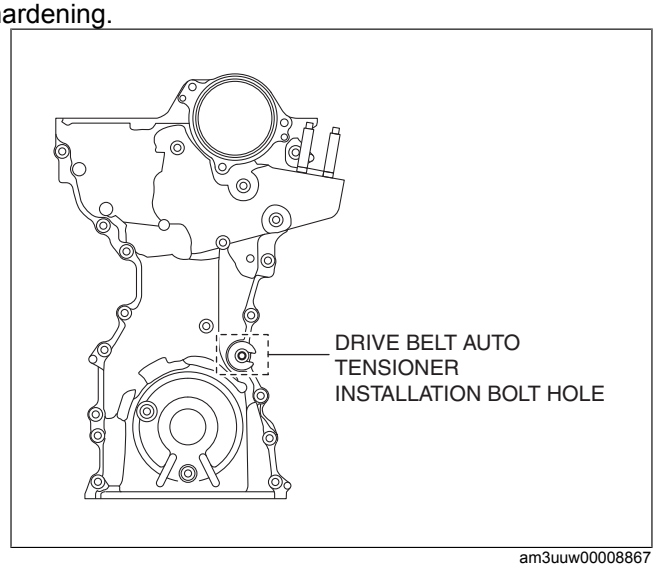
6. Install the engine front cover to the engine.

Note
- Temporarily install an appropriate bolt to the drive belt auto tensioner installation bolt hole to prevent:
  - A silicone sealant adhesion malfunction in the drive belt auto tensioner installation bolt hole.
  - A bolt mis-installation due to silicone sealant hardening.

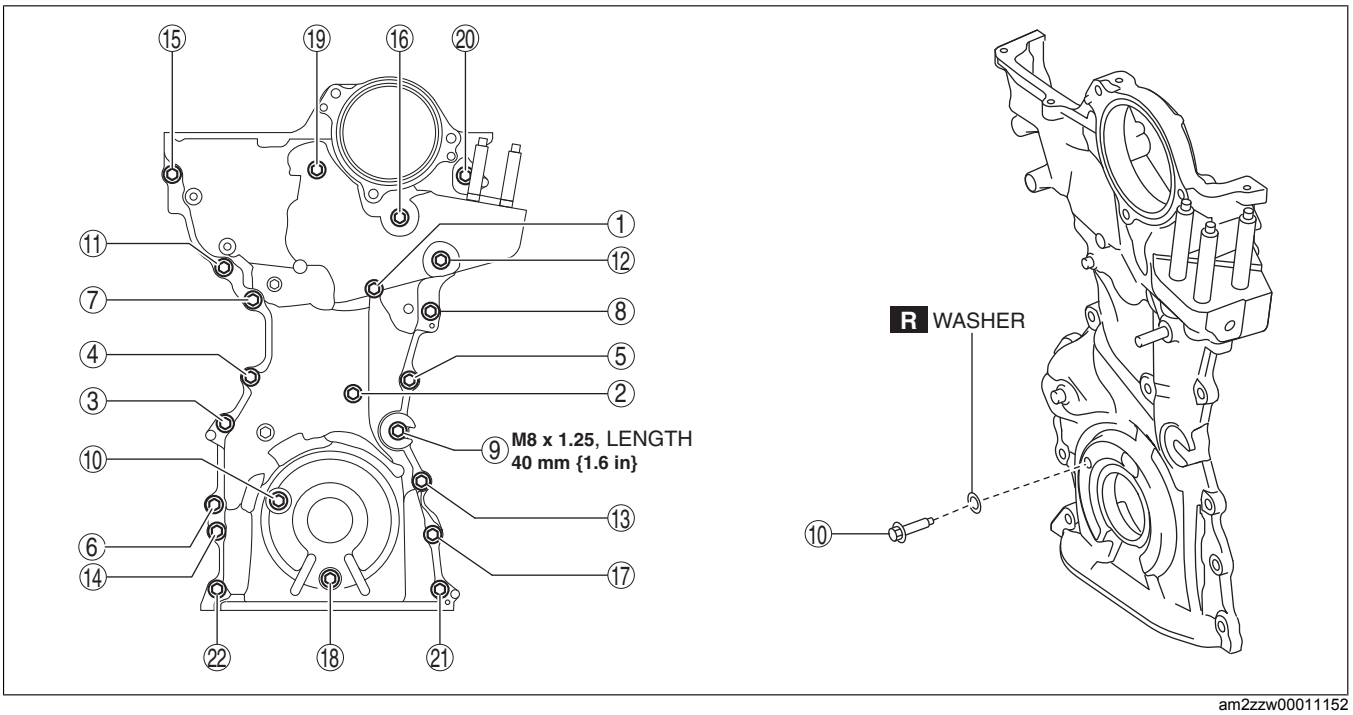
7. Prepare an appropriate **M8 × 1.25** bolt (length **40 mm {1.6 in}**).

Caution
- **For the number 10 bolt of the tightening order, install the bolt with new washer.**

8. Tighten the engine front cover installation bolts in the order shown in the figure.



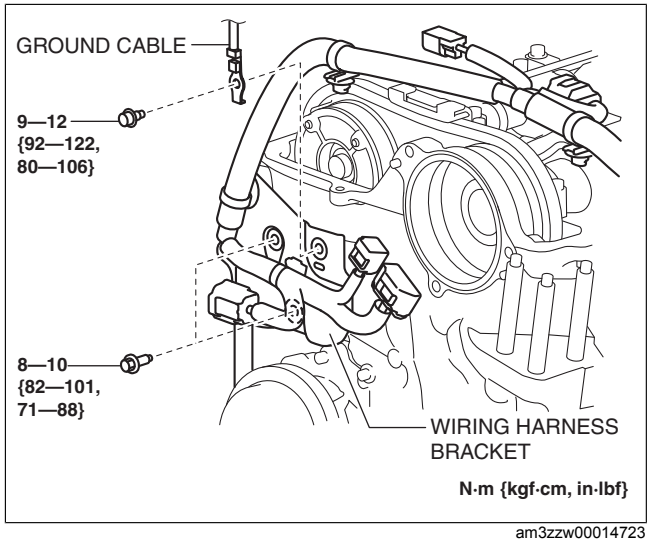
am3uuw00008867



am2zzw00011152

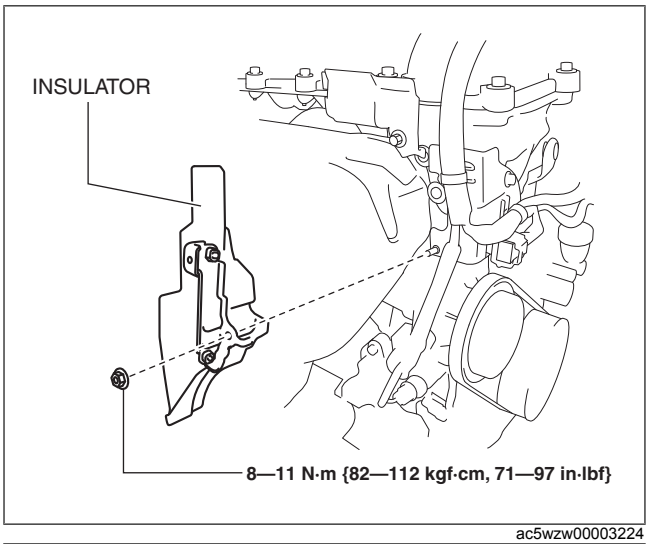
Tightening torque
20—26 N·m {2.1—2.6 kgf·m, 15—19 ft·lbf}

9. Remove the bolt installed to the drive belt auto tensioner installation bolt hole when installing the drive belt auto tensioner.
10. Install the ground cable and wiring harness bracket shown in the figure.



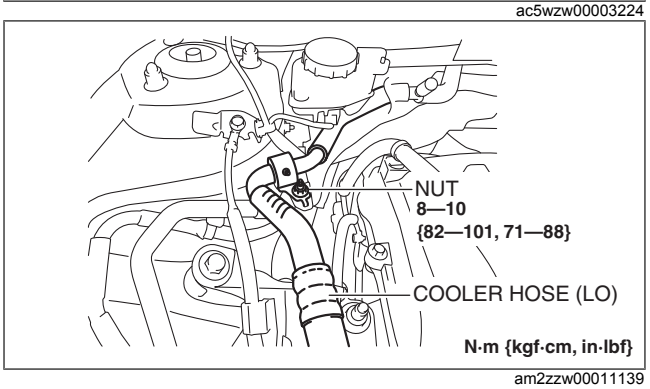
am3zzw00014723

11. Install the insulator shown in the figure.



ac5wzw00003224

12. Install the cooler hose (LO).



am2zzw00011139

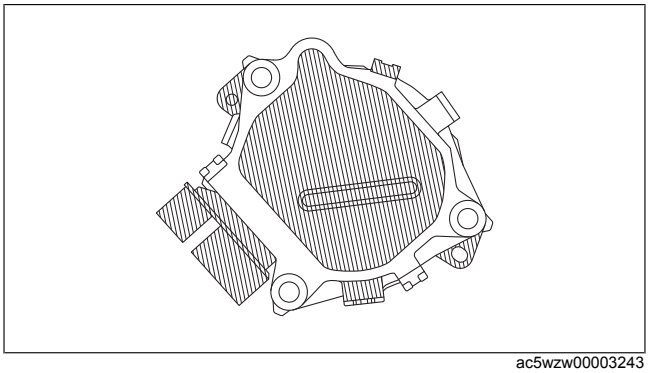
Electric variable valve timing motor/driver installation note

1. Install a new O-ring to the O-ring installation groove of the engine front cover.

Caution

- **To prevent damage to the electric variable valve timing motor/driver, do not apply excessive force (force of 100 N {10.2 kgf, 22.5 lbf} or more) to the shaded areas shown in the figure.**

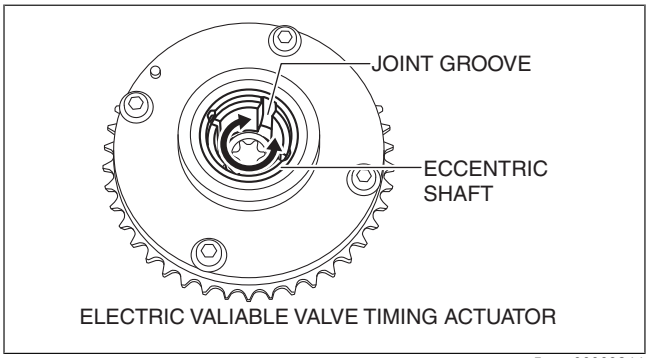
2. Install the electric variable valve timing motor/driver using the following procedures.



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Note

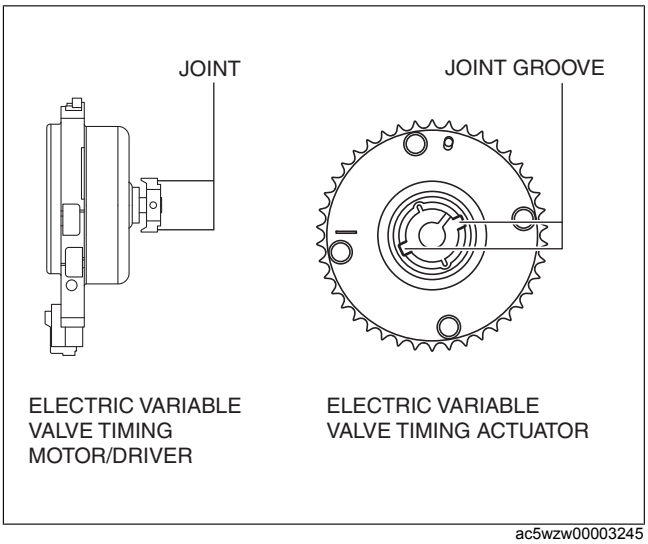
- The eccentric shaft on the electric variable valve timing actuator side can be rotated to the left and right.
- The electric variable valve timing motor/driver can be assembled with the joint groove of the eccentric shaft in any position, and it will not lead to vehicle damage or performance reduction.



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(1) Before installation, rotate the joint on the end of the electric variable valve timing motor so that it is aligned to the joint groove on the electric variable valve timing actuator side.
(2) Engage the joint on the end of the electric variable valve timing motor with the joint groove on the electric variable valve timing actuator side.
(3) Attach the seal surface.
(4) Tighten the electric variable valve timing motor/driver installation bolts.

Tightening torque
20—26 N·m {2.1—2.6 kgf·m, 15—19 ft·lbf}



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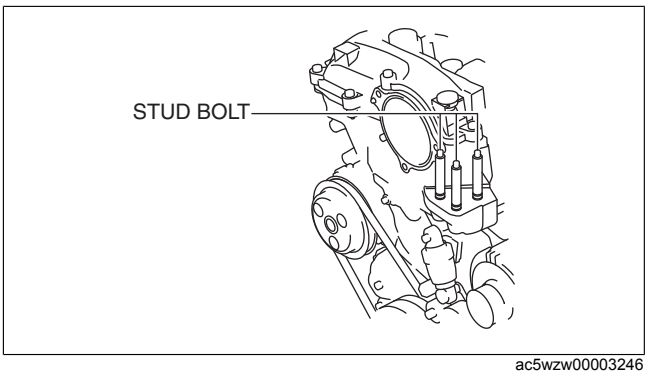
No.3 engine mount installation note

Caution
- **If the No.3 engine mount is deviated from the original position when installing the No.3 engine mount, engine noise or vibration could increase. When installing the No.3 engine mount, align the alignment mark placed during removal and install it to the original position.**

Note
- When replacing the No.3 engine mount, place a mark at the same position as the one placed before removal.
- If the No.3 engine mount nut is loosened, tighten the engine front cover stud bolts because they may have loosened.

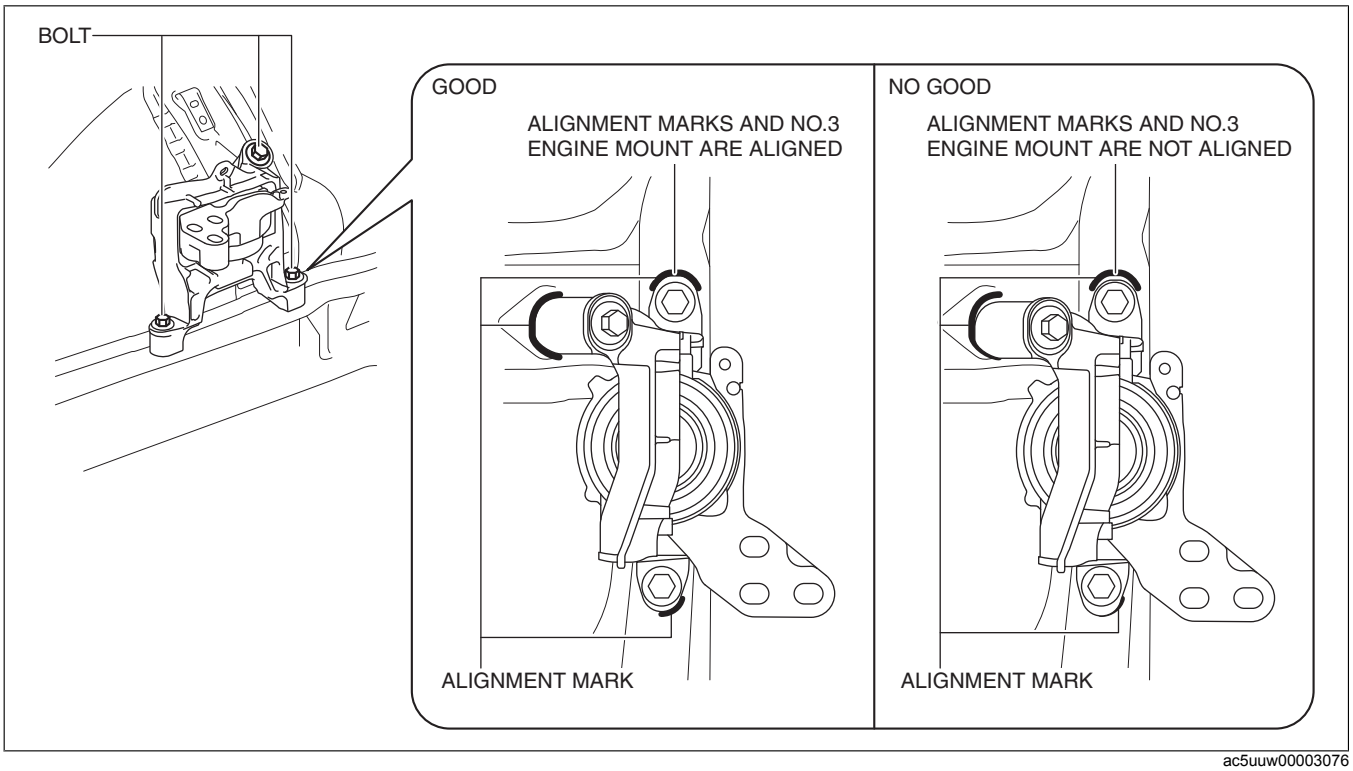
1. Tighten the engine front cover stud bolts.

Tightening torque
7—13 N·m {72—132 kgf·cm, 62—115 in·lbf}



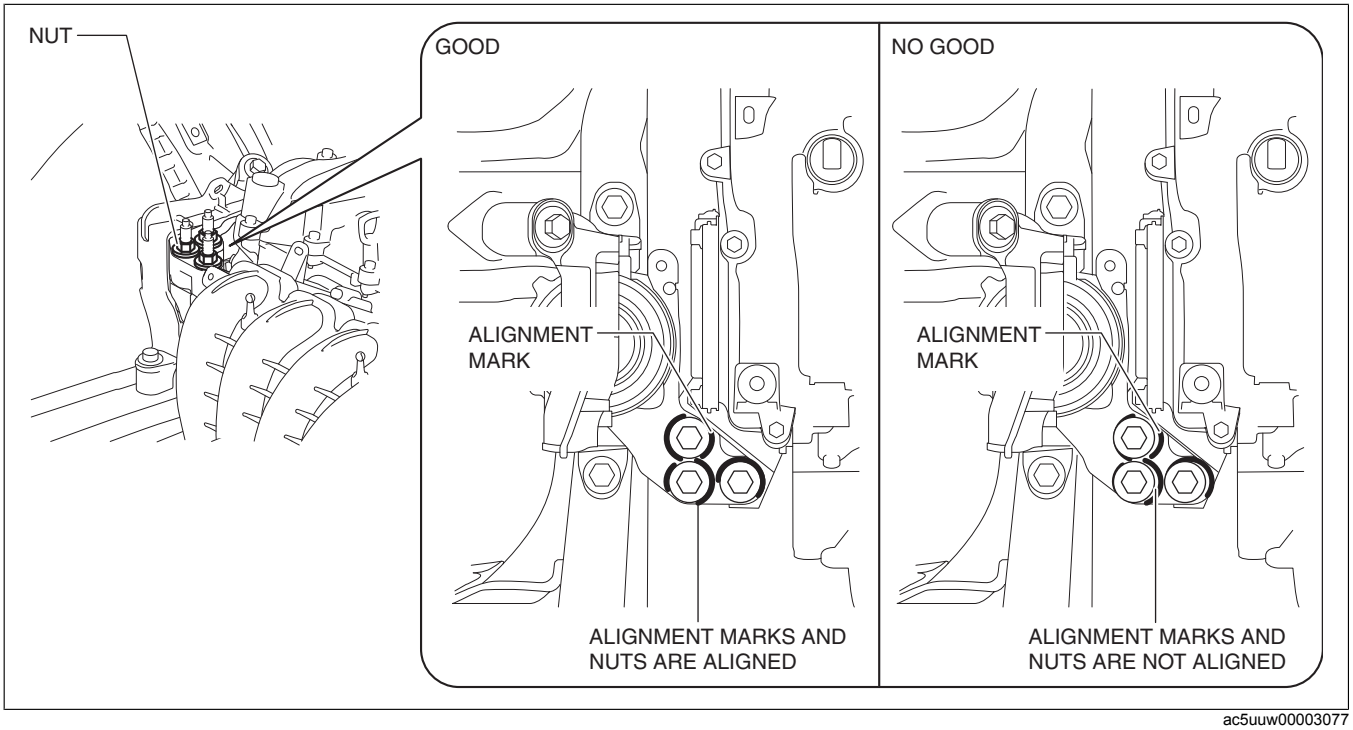
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2. Temporarily tighten the No.3 engine mount installation bolts and nuts using the following procedure:
   (1) Align the alignment marks on the body side and No.3 engine mount, and temporarily tighten the bolts shown in the figure.



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(2) Temporarily tighten the nuts shown in the figure while aligning the alignment marks of the No.3 engine mount and nuts.



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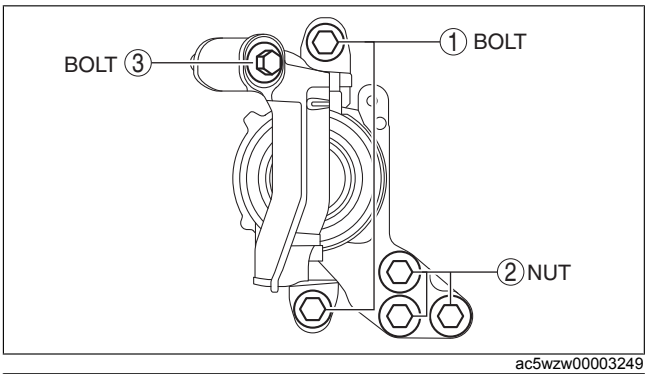
Note

- If the alignment marks are not aligned, align the alignment marks while slightly moving the engine and temporarily tighten the nuts.

3. Tighten the No.3 engine mount installation bolts and nuts in the order as shown in the figure.

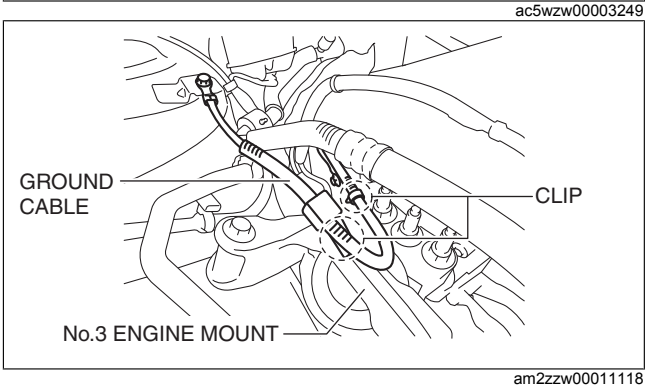
Tightening torque

Installation position	Tightening torque
1	76—95 N·m {7.8—9.6 kgf·m, 57—70 ft·lbf}
2	82—95 N·m {8.4—9.6 kgf·m, 61—70 ft·lbf}
3	49—65 N·m {5.0—6.6 kgf·m, 37—47 ft·lbf}



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4. Remove the **SST**.
5. Install the clip of ground cable to the No.3 engine mount.



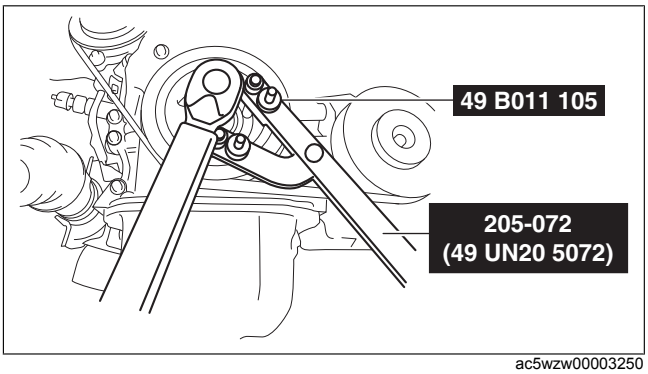
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Crankshaft pulley lock bolt installation note

Caution
- **To prevent damage to the crankshaft pulley, protect the crankshaft pulley with a clean cloth so that the bar part (205-072) of the SST does not contact the crankshaft pulley.**
- **To prevent damage to the engine front cover, be careful not to allow the tab part (49 B011 105) of the SST to get caught between the crankshaft pulley and the engine front cover.**

1. Install the **SST** on the crankshaft pulley to lock the crankshaft against rotation.
2. Tighten the crankshaft pulley lock bolt in the order shown in the following two steps.

Tightening procedure
Step 1: 90—110 N·m {9.2—11 kgf·m, 67—81 ft·lbf}
Step 2: 55—65°



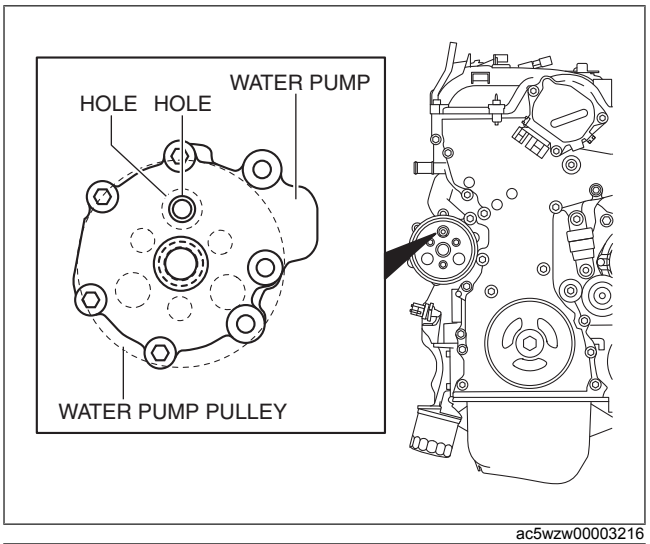
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Water pump pulley installation note

Caution
- **Be careful not to damage the belt groove and surface of the water pump pulley when using tools, otherwise it will cause wear, breakage, abnormal noise of the drive belt (stretch belt), damage to the pulley, and rust.**

1. Install the water pump pulley to the water pump and temporarily tighten the bolt.

2. Align the water pump pulley hole with the water pump hole as shown in the figure.

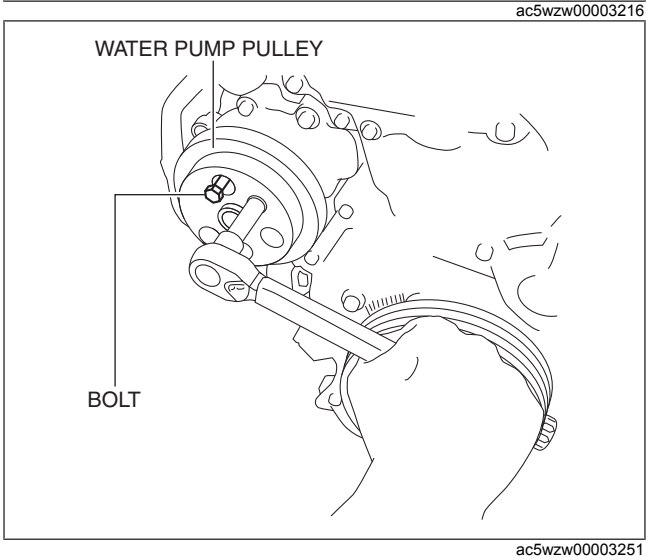


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3. Insert an appropriate bolt (length **70 mm {2.8 in}**) into the water pump hole shown in the figure and lock the water pump pulley against rotation.
4. Completely tighten the water pump pulley bolt to the specified torque.

   Tightening torque
   8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}

5. Remove the bolt used for locking the water pump pulley against rotation.



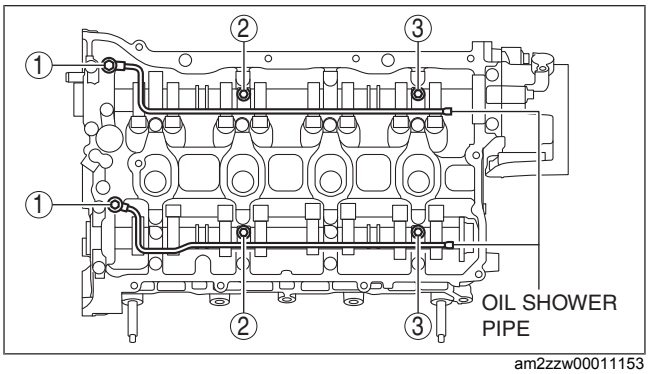
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Oil shower pipe installation note

1. Install the oil shower pipe in the order shown in the figure.

Tightening torque

Installation position	Tightening torque
1	9—12 N·m {92—122 kgf·cm, 80—106 in·lbf}
2, 3	8.0—9.0 N·m {82—91 kgf·cm, 71—79 in·lbf}

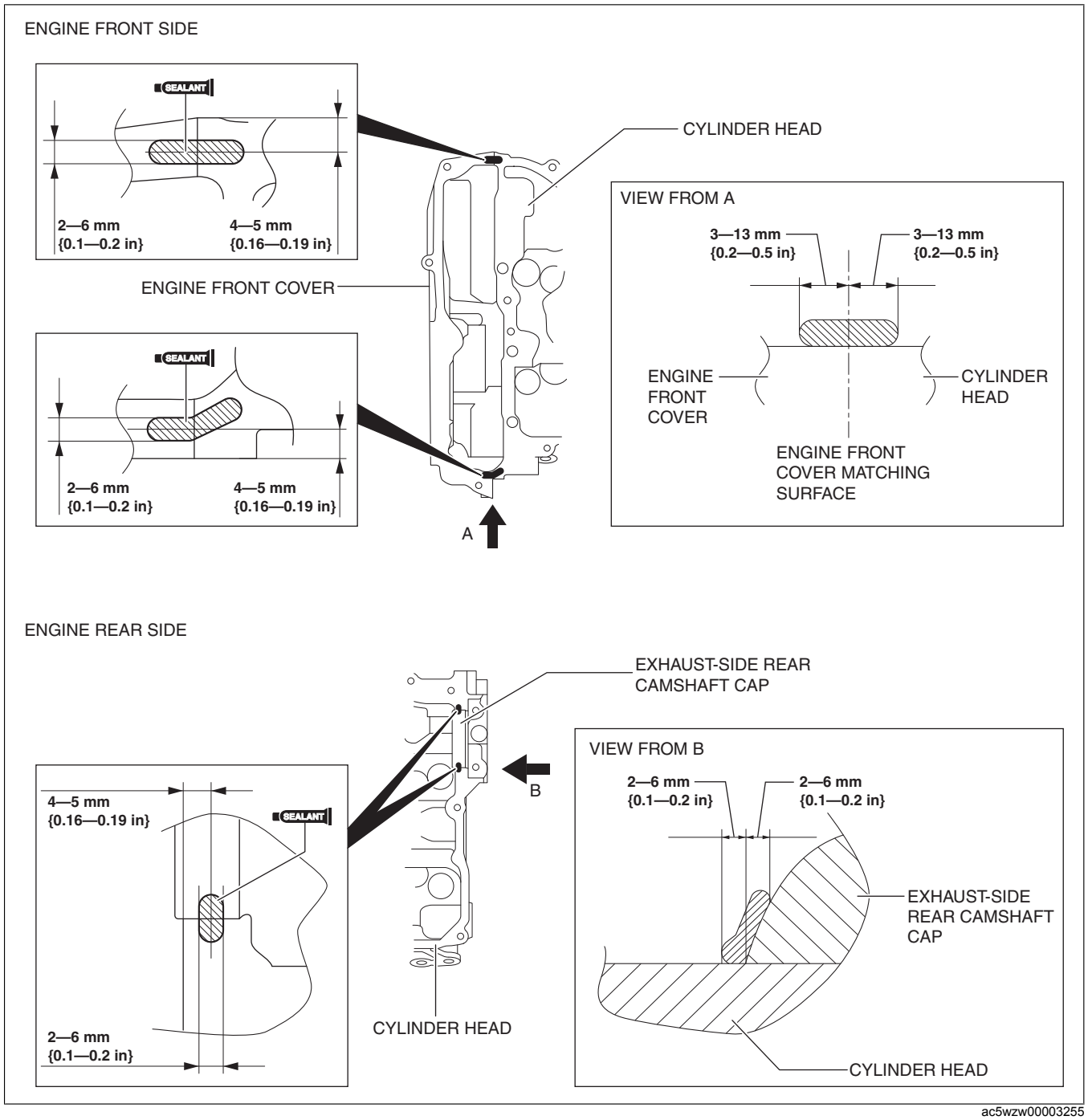


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Cylinder head cover installation note

Caution
- **To assure the sealing performance of the cylinder head cover, be careful of the following:**
  - **— Verify that the cylinder head cover gasket is inserted into the cylinder head cover groove and install the cylinder head cover.**
  - **— Completely clean and remove oil, dirt, sealant or other foreign matter from the seal surface.**
- **To prevent silicone sealant from hardening, adhere the cylinder head cover and the cylinder head firmly within 10 min. after applying silicone sealant. After adhering them, tighten the installation bolts immediately.**

1. Insert a new cylinder head cover gasket into the cylinder head cover groove.
2. Apply silicone sealant (TB1217D or equivalent) to the areas shown in the figure.

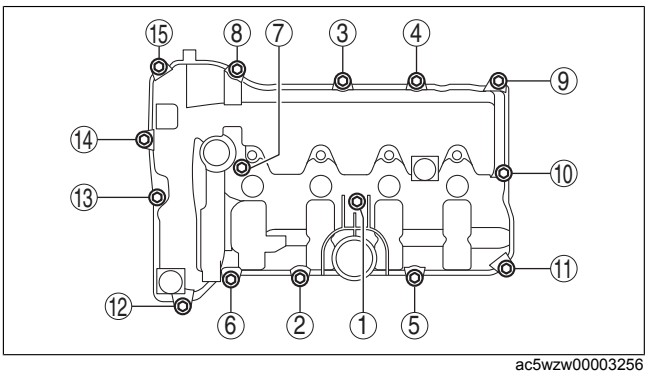


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3. Install the cylinder head cover.
4. Temporarily tighten the cylinder head cover bolts.
5. Tighten the cylinder cover bolts in the order shown in the figure.

   Tightening torque
   6.0—7.0 N·m {62—71 kgf·cm, 54—61 in·lbf}

6. Measure the tightening torque again and verify that it is **6.0 N·m {61 kgf·cm, 53 in·lbf}** or more.
7. Connect the vacuum hose.
8. Connect the ventilation hose.
9. Return the wiring harness which was set away from the cylinder head cover and the intake manifold to its original position.
10. Connect the OCV connector.



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